MILITARY SPECIFICATION

COATS, AMMUNITION CARRYING, BUOYANT, AND BLADDER, FLOTATION

1. SCOPE

1.1 Scope.- This specification covers the requirements for buoyant ammunition carrying coats fabricated of camouflage rip-stop cotton cloth; and inflatable bladders, made of polychloroprene coated nylon, used with the coats to provide flotation.

1.2 Classification

1.2.1 Coats.- The coats shall be of three types and in the sizes as follows (see 6.2):

Types

Type I - Rifleman
Type II - Grenadier
Type III - Radioman

Sizes

Small - regular	Medium - short	Large - short
Small - long	Medium - regular	Large - regular
	Medium - long	Large - long

1.2.2 Bladders.- The bladders shall be of one size and in the following styles:

Styles

Style I - Rifleman, Grenadier
Style II - Radioman

2. APPLICABLE DOCUMENTS

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2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip (Polyolefin)
V-B-871	- Button, Sewing Hole, and Button, Staple (Plastic)
V-T-276	- Thread, Cotton
V-T-280	- Thread, Cotton Gimp, Buttonhole
V-T-295	- Thread, Nylon
JJ-W-155	- Webbing, Textile (Cotton, Elastic)
QQ-W-343	- Wire, Electrical (Uninsulated)
UU-P-268	- Paper, Kraft, Untreated, Wrapping
DDD-L-20	- Label, For Clothing, Equipage, and Tentage, (General Use)
DDD-T-86	- Tape, Textile, Cotton, General Purpose (Unbleached, Bleached, or Dyed)
PPP-B-636	- Box, Fiberboard
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-B-371	- Braid, Textile, Tubular
MIL-T-5038	- Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-C-5040	- Cord, Nylon
MIL-A-5540	- Adhesive, Polychloroprene
MIL-P-5516	- Packing, Preformed, Petroleum Hydraulic Fluid Resistant, 160°F
MIL-L-15581	- Life Preservers, CO_2 Inflatable, Pouch Type
MIL-G-16491	- Grommet, Metallic
MIL-B-17757	- Boxes, Fiberboard, Corrugated (Modular Sizes)
MIL-C-19002	- Cloth, Coated, and Tape, Coated Cloth - Polychloroprene on Nylon, Pneumatic Life Preserver
MIL-F-21840	- Fastener Tape, Hook and Pile, Nylon
MIL-C-25369	- Cylinder, Carbon Dioxide, for Inflation of Pneumatic Life Preservers
MIL-C-43303	- Cord, Elastic, Cotton
MIL-C-43468	- Cloth, Camouflage Pattern, Wind Resistant Poplin, Cotton
MIL-C-46842	- Cushioning Material, Unicellular Polyethylene Foam (For Packaging Purposes)
MIL-O-81375	- Oral Inflation Assemblies, Survival Equipment, Inflatable
MIL-M-81385	- Manifold Stem Assemblies, Life Preserver

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STANDARDS

FEDERAL

FED-STD-601 - Rubber, Sampling and Testing
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspections by Attributes
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

BUREAU OF SHIPS. No. 608-1895224 - Pressure Relief Valve
NAVSHIPS No. 805-1411524; CO_2 Holder and Holder Gauge

(Copies of specifications, drawings, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606).

3. REQUIREMENTS

3.1 Guide sample.- Samples, when furnished, are solely for guidance and information to the supplier (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

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3.2 First article approval.- The requirement for first article shall be as specified by the procuring activity (see 6.2).

3.3 Coat.-

3.3.1 Materials.-

3.3.1.1 Basic material.- The basic material shall be wind resistant cotton poplin camouflage cloth conforming to type I of MIL-C-43468.

3.3.1.2 Thread.- The thread for seaming and stitching the coat shall be cotton shade olive drab S-1, cable No. 66022 conforming to type IA3, 3-ply, ticket Nos. 30 and 50; and 2-ply, ticket No. 70 of V-T-276.

3.3.1.2.1 Colorfastness.- The dyed thread shall show fastness to light, and combined laundering and chlorine bleaching equal to or better than the standard sample when tested as specified in V-T-276. When no standard sample is available, the dyed thread shall show a minimum of "good" fastness to light, and combined laundering and chlorine bleaching.

3.3.1.3 Gimp.- The cotton gimp for reinforcing buttonholes shall be size No. 8, shade olive drab S-1, cable No. 66022 conforming to type I or II of V-T-280.

3.3.1.3.1 Colorfastness.- The dyed gimp shall show fastness to laundering and chlorine bleaching equal to or better than the standard sample when tested as specified in V-T-280. When no standard sample is available, the dyed thread shall show a minimum of "good" fastness to laundering and chlorine bleaching.

3.3.1.4 Buttons.- The buttons shall be plastic conforming to type II, class D, DOD shade BP, cable No. 62016, dull finish, style 26, size 30-line of V-B-871.

3.3.1.5 Fastener tapes, nylon.- The matching hook and pile nylon fastener tapes shall be one inch conforming to type II, class 1 of MIL-F-21840. The color shall be olive green.

3.3.1.6 Cord, elastic.- The elastic cord for the coat hems shall conform to class 1, olive green, of MIL-C-43303.

3.3.1.7 Cushioning.- The cushioning material for the protection pad shall be polyethylene 1/4 inch thick, natural, conforming to MIL-C-46842.

3.3.1.8 Braid.- The braid for securing the first aid kit to the coat shall be olive green conforming to type IV, class 1 of MIL-B-371.

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3.3.1.9 Webbing, elastic.- The elastic webbing for pocket restraint straps shall be 1/2 inch wide, olive green, conforming to Type 2, class 3 of JJ-W-155.

3.3.1.10 Webbing, nylon.- The webbing for the pocket straps and bladder restraint straps shall be olive green, 1 inch wide, conforming to type IV of MIL-T-5038.

3.3.1.11 Sound Separators.- The sound separators shall be a thermo 1/8 inch Curon 508-C charcoal polyurethane foam laminated on each side to a 15 denier black tricot knit nylon, conforming to style IS-9-017 manufactured by Reeves Bros., 517 Lyons Avenue, Irvington, New Jersey, or a functionally interchangeable equivalent.

3.3.1.12 Grommets.- The grommets and washers for the coat hem shall be brass conforming to type I, size No. 0, class 3 of MIL-G-16491.

3.3.1.13 Identification and instruction label.- Each coat shall have a combination identification and instruction label conforming to type I, classes 1 and 3 of DDD-L-20. The printing shall show fastness to laundering. The label shall bear the following inscription:

COAT, BUOYANT, AMMUNITION CARRYING
TYPE I - RIFLEMAN (Example)
Name of Supplier:
Contract No.: DSA-000-000-000 (Example)
Stock No.: 8415-000-0000 (Example)
Size: Medium - Regular (Example)

INSTRUCTIONS: For buoyancy, the inflatable bladder, flotation provided separately should be inserted before use. The bladder shall be inserted through either back pocket opening or the opening at back right waist line.

3.3.1.14 Adhesive tickets.- Adhesive tickets, when used for size markings, shall be of paper with a thermo-activated adhesive applied on one side that will withstand moderate steam pressing operations. The adhesive shall not discolor the material and the adhesive mass shall not adhere to the material upon removal of the tickets.

3.3.1.15 Tape, textile, cotton.- The tape for reinforcing the back pouch in the type III coat shall be 5/8 inch in width, conforming to type I or II, class 1 or 2 of DDD-T-86.

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3.3.1.16 Resin impregnating agent.- The resin agent used for impregnating or dipping the ends of cotton braid and webbing shall be cellulose acetate or cellulose acetate butyrate.

3.3.2 Design.- The coats shall be single breasted design with buttoned front closure, fabricated of a cotton poplin with camouflage pattern, with straight cut bottom and long buttoned cuffed sleeves. The grenadier and radioman coats shall have a body and an outershell consisting of a full back, a full front, and one-half front and one-half back joined at the body seams. The rifleman coat shall have a body and an outershell consisting of a full front, a one-half front and two full backs joined at the body seams. A combination of bellows-style pockets with buttoned flaps, ammunition pouch-style pockets with flaps, and compass-style pockets with flaps, both secured by nylon tape fasteners, shall be attached to the coats. Each coat shall have belt loops and a tie-cord shall be threaded through grommets into a tunnel-hem at the bottom.

3.3.2.1 Figures.- Figures 1 thru 4 are furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figures, the written specification shall govern.

3.3.3 Patterns.- Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The patterns allow 1/4 to 3/8 inch for all seams except where otherwise indicated in the tables of operation. Double-lapped double-stitched seams shall be 1/4 to 5/16 inch gauge. The standard patterns shall not be altered in any way and are to be used for cutting the supplier's working patterns. The working patterns shall be identical to the standard patterns.

3.3.3.1 Pattern parts.- The component parts shall be cut from the material specified in accordance with the following pattern parts:

Pattern parts	Cut parts		
	Rifleman	Grenadier	Radioman
Cloth, camouflage			
Front	2	2	2
Back	2	1	1
Top back	-	1	1
Top,front	1 (L)	2 (L&R)	1 (L)
Top front	1 (R)	-	1 (R)
Undersleeve	2	2	2
Topsleeve	2	2	2
Collar	2	2	2

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Pattern parts	Cut parts		
	Rifleman	Grenadier	Radioman
Cloth, camouflage (cont'd)			
Cuffs	2	2	2
Compass pocket	2	2	2
Protective flap	2	2	2
M-16 front pouch reinforcement piece	2	-	*
M-16 back pouch reinforcement piece	4	-	-
M-16 magazine pouch	8	4	6
M-16 magazine pouch flap	8	4	6
M-16 magazine pouch flap facing	8	4	6
Back pocket opening flap	2	-	-
Back pocket opening facing	2	-	-
Pencil pocket	1	1	1
Tab holder (first aid pouch)	1	1	-
Bellow pocket	2	2	2
Inside bellows pocket	2	2	2
Outside bellows pocket	2	2	2
Flap bellows pocket	2	2	2
40 MM grenade pouch	-	7	-
40 MM grenade pouch flap facing	-	7	-
Back pouch	-	-	2
40 MM grenade back pouch reinforcement piece	-	1	-
Reinforcement piece (grenade and magazine)	-	6	-
Valve opening facing	1	1	1
Oral tube opening facing	1	1	1
Tie cord reinforcement (first aid pouch)	1	1	-
Loops	**	**	**
Cushioning			
Protective flap interlining	1	1	1
Polyurethane foam			
Sound separator	8	4	6

* Single pouches reinforced with 2 3/4 inch by 3 1/2 inch pieces of material.
** As specified in construction table.

3.3.4 Construction.-

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3.3.4.1 Stitch, seam, and stitching types.- Stitch, seam, and stitching types specified in Tables I, II and III shall conform to FED-STD-751. Wherever two or more methods for seam or stitch types are given for the same part of the operation, any one of them may be used. Where stitch type 401 is used, the looper or underthread shall be on the inside of the coat. Seam allowances shall be maintained with seams properly sewn so that no raw edges, runoffs, twists, pleats or open seams will result. Unless otherwise specified, edge-stitching or outer rows of stitching shall be 1/16 inch from the edge. Unless otherwise specified, tolerances shall be ± 1/16 inch.

3.3.4.2 Thread breaks and ends of seams.- Ends of all seams and stitchings shall be backtacked not less than 1/4 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or 401 stitch type thread breaks may be repaired by using 301 stitch type.

3.3.4.3 Stitches per inch.- The minimum and maximum number of stitches per inch shall be as specified in Tables I, II and III.

3.3.4.4 Buttonholes.- The buttonholes shall be eyelet-end, taper-bar, cut-after type, worked over gimp, with the purling on the outside, and the end of the buttonhole stitching and gimp securely tacked. The bar at the end of the taper bar shall have a minimum of four tacks not including the first cross-over stitch.

3.3.5 Manufacturing operations requirements.- The coats shall be manufactured in accordance with the operation requirements specified in Tables I, II and III. The supplier is not required to follow the exact sequence of operations listed provided that the finished coats are identical to those produced by following the sequence of operations as listed in Tables I, II and III.

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Table I - Construction of Rifleman (Type I) Coat

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
	Rifleman (Type I) Coat					
1.	Cutting a. Cut the coats in strict accordance with patterns, which show size, shape, placement of pockets and flaps, warp directional lines, and notches for the proper assembling of all parts.					
2.	Replacement of damaged parts a. Replace at time of cutting any part containing damages such as a hole, smash, multiple float or slub. b. Replace any part damaged during the manufacturing process by a needle chew, cut, tear, hole, mend, burn or exposed drill hole.					
3.	Marking.- Mark or ticket cut parts of the coat to insure a uniform size and proper assembly throughout the coat.					
4.	Make M-16 magazine pouches a. Overedge sound separator on all sides.	502 or 503	EFd-1	10-14	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Make M-16 magazine pouches (cont'd)					
	b. Mark and stitch four (4) 3/16 inch (inside diameter) eyelets on each M-16 magazine pouch piece as indicated by marks on pattern. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
	c. Cut two 12 (± 1/4) inch pieces of nylon webbing. Fuse ends.					
	d. Hem top of pouch as indicated by marks on pattern. Fold under raw edge. Stitch 1/16 to 1/8 inch from fold, simultaneously inserting the strip of nylon webbing in the hem centered with end adjacent to fold of hem.	301	SSl-1 and EFb-1	10-14	50/3	50/3
	e. Fold pouch face to face and stitch bottom sides 1/4 (± 1/16) inch from raw edge.	301	SSa-1	10-14	50/3	50/3
	f. Turn to finished position. Fold magazine pouch piece as indicated on patterns and topstitch 1/16 inch from edge.	301	OSf-1	10-14	50/3	50/3
	g. Position a 1 inch length of pile tape centered 3/4 inch from the top of the pouch hem and stitch 1/16 inch from edge all sides.	301	LSbj-1 all sides	10-14	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Make M-16 magazine pouches (cont'd)					
	h. Center a 2 1/2 inch length of 1/2 inch wide elastic webbing on the nylon webbing 3 1/2 inches from the stitched end so that the elastic webbing faces the back of the pouch material. Stitch and backstitch 1/8 to 1/4 inch from end.	301	LSbj-1	10-14	50/3	50/3
	i. Stitch sound separator to inside of pouch 3 sides catching 1/8 to 1/4 inch of the free end of the elastic webbing centered in the bottom of the pouch.	301	SSb-1	10-14	50/3	50/3
	j. Join face to face flap facing to pouch flap simultaneously closing sides and bottom of pouch and inserting and catching 1/4 to 3/8 inch a 2 1/2 inch length of elastic webbing in the center bottom.	301	SSa-1 and SSe-2(a)	10-14	50/3	50/3
	k. Turn flap and pouch to finished position. Topstitch flap 1/16 inch from edge 3 sides.	301	SSe-2(b)	10-14	50/3	50/3
	l. Center the end of a 12 inch piece of nylon webbing under the remaining side of the flap facing approximately 1/2 inch. Stitch side of flap to pouch 1/16 inch from edge.	301	LSd-1 and Similar to EFf-1	10-14	50/3	50/3
	m. Stitch 1/8 to 1/4 inch from edge free end of elastic webbing to underside of nylon webbing at a distance of 3 inches from stitched end of the nylon webbing.	301	SSa-1	10-14	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Make M-16 magazine pouches (cont'd)					
	n. Center a 1 inch length of hook tape on the flap facing 1/4 inch from the edge. Stitch and backstitch a 1/2 by 7/8 inch rectangle through fastener and flap with free end of the fastener towards outer portion of flap.	301	LSbj-1	10-14	50/3	50/3
5.	Attach front M-16 magazine pouches					
	a. Stitch M-16 front pouch reinforcement pieces to top fronts as indicated by marks on patterns; 1/16 inch from edge, all sides.	301	LSbj-1 each side	10-14	50/3	50/3
	b. Position M-16 pouches on top fronts as indicated by marks on patterns with flap open as follows: 3 on left side 1 on right side					
	c. Stitch pouches to fronts on all sides, 1/16 inch from edge, 3 sides and across top of pouch through flap.	301	LSd-1 (sides) SSv-1 (top)	10-14	50/3	50/3
	d. Place two 1/2 inch vertical bartacks at the tops of each pouch at hem, 1/16 inch from sides.	Bartacks		28 per bartack	50/3	50/3
6.	Attach webbing restraint loops					
	a. Cut two 2 (± 1/4) inch lengths of nylon webbing. Fuse ends.					

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
6.	Attach webbing restraint loops (cont'd)					
	b. Fold under each end 1/4 inch. Position on face side of fronts as indicated by marks on patterns and stitch 1/16 inch from folded edges.	301	LSd-1 each end	10-14	50/3	50/3
7.	Prepare belt loops					
	a. Cut stripping for belt loops of sufficient width to finish 5/8 to 3/4 inch wide.					
	b. Fold stripping in half lengthwise, turn in raw edges and stitch 1/16 to 1/8 inch from each edge.	301	EFp-2	10-14	50/3	50/3
	c. If one side of stripping is selvaged, fold stripping lengthwise, turn in raw edge and stitch 1/16 to 1/8 inch from each edge.	301	EFz-2	10-14	50/3	50/3
8.	Attach front loops Position top edge of belt loop on right and left top front as indicated by marks on patterns. Fold ends under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from fold.	Bartack	LSd-1	28 per bartack	50/3	50/3
9.	Make left top front facing Fold left top front as indicated by marks on pattern to form front facing and stitch 1/4 inch from fold.	301	OSf-1	10-14	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
10.	Attach back M-16 magazine pouches					
	a. Stitch M-16 back reinforcement pieces to back as indicated by marks on pattern, 1/16 inch from edge, all sides.	301	LSbj-1 each edge	10-14	50/3	50/3
	b. Position four M-16 pouches with flaps open on backs as indicated by marks on pattern.					
	c. Stitch pouches on all sides, 1/16 inch from edge 3 sides and across top of pouch through flap.	301	LSbj-1 3 sides SSv-1 top	10-14	50/3	50/3
	d. Place two 1/2 inch vertical bartacks at the tops of each pouch at hem.	Bartack		28 per bartack	50/3	50/3
11.	Complete outer backs					
	a. Stitch first aid kit reinforcement piece to back side of back as indicated by marks on pattern, 1/16 inch from edge, all sides.	301	LSbj-1 each edge	10-14	50/3	50/3
	b. Mark and stitch two (2) 3/16 inch (inside diameter) eyelets through back and reinforcement piece as indicated by marks on pattern. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
	c. Cut an 18 (± 1/2) inch piece of cotton braid. Dip ends to prevent raveling as specified in 3.3.1.16.					
	d. Thread braid through eyelets with ends on the outside of back.					

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
11.	Complete outer backs (cont'd)					
	e. Place a 1/2 inch vertical bartack centrally between the eyelets, securing the braid.	Bartack		28 per bartack	50/3	50/3
	f. Fold first aid tab holder in half face to face lengthwise. Stitch sides 1/4 inch from edges.	301	SSa-1	10-14	50/3	50/3
	g. Trim points and turn face out.					
	h. Fold under raw edges 1/4 inch and stitch 1/16 inch from edge.	301	SSc-1	10-14	50/3	50/3
	i. Mark and stitch two (2) 3/16 inch (inside diameter) eyelets as indicated by marks on pattern. Purling shall be facing the outside.	Eyelet		22 per eyelet	50/3	50/3
	j. Fold lengthwise with eyelets on inside and stitch ends 1/8 inch from edges securely backtacking the ends.	301	SSa-1	10-14	50/3	50/3
	k. With eyelets on the lower ply, place tab centered 1/2 inch below top edge of back and stitch 1/16 inch from edge, continuing halfway down both sides.	301	Similar to LSd-1	10-14	50/3	50/3
12.	Make lower back opening					
	a. Fold lower back opening flap in half lengthwise face to face and stitch 1/4 to 3/16 inch gauge each end. Turn to finished position and edge stitch bottom and sides 1/4 to 3/16 inch from edges.	301	SSe-2 and OSf-1	10-14	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
12.	Make lower back opening (cont'd)					
	b. Make a 7/8 inch cut length vertical buttonhole on the flap as indicated by marks on pattern with the inside edge of the eyelet 5/8 to 3/4 inch from the folded edge.	Button-hole		52-56 per buttonhole	30/3 or 50/3	50/3
	c. Position face side of pocket facing opening piece on face side of back sandwiching flap in between on the upper side.					
	d. Stitch a rectangular box pattern through all plies as indicated by marks on patterns.	301	LSbj-1 each edge	10-14	50/3	50/3
	e. Slit center, notching ends.					
	f. Turn facing to inside and edgestitch 1/16 inch from folded edge all sides, with flap down, forming a rectangular opening.	301	SSe-2(b)	10-14	50/3	50/3
	g. Turn under raw ends of facing and stitch 1/16 inch from folded edge all sides.	301	LSd-1	10-14	50/3	50/3
	h. Place a 1/2 inch vertical bartack at ends of pocket opening.	Bartack		28 per bartack	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
13.	Position back belt loops a. Position three belt loops on backs as indicated by marks on patterns. Fold ends under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from each fold. b. Finished loop openings shall measure 3 1/2 (± 1/8) inches.	Bartack	LSd-1	28 per bartack	50/3	50/3
14.	Join shoulders Join, face to face, top fronts and fronts to backs at shoulders with double-lapped, double stitched seams 1/4 to 5/16 inch gauge. Finished seams shall face back for outer shell and face front for inner shell.	301 or 401	LSc-2	10-14	50/3 50/3	50/3 70/2
15.	Make bellows pockets a. Mark and stitch two (2) 3/16 inch (inside diameter) eyelets on the inside bellows lower pocket, as indicated by marks on pattern. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
	b. Fold pocket in half lengthwise face to face, and stitch, from top raw edge down to 5 (± 1/8) inches.	301	SSa-1	10-14	50/3	50/3
	c. Form an inverted box pleat (on outside of pocket) and stitch across bottom of pleat to hold pleat in place.	301		10-14	50/3	50/3
	d. Hem top of pocket as indicated by marks on pattern and stitch 1/16 to 1/8 inch from edge with turnunder 1/4 to 3/8 inch.	301	EFb-1	10-14	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
15.	Make bellows pockets (cont'd)					
	e. Stitch inside and outside bellows together face to face, 1/4 inch from the edge.	301	SSa-1	10-14	50/3	50/3
	f. Stitch pockets to bellows 1/4 inch from the edge.	301	SSe-2(a)	10-14	50/3	50/3
	g. Hem top of bellows with the raw edge turned in and the seam allowance of the pocket, turned toward bellows caught in the hem. The top edge of bellows shall be even with the top of pocket.	301	EFb-1	10-14	50/3	50/3
	h. Edge stitch pocket 1/16 to 1/8 inch from edge all three sides. The ends of the stitching at the top of the bellows shall be backtacked not less than 1/2 inch.	301	SSe-2(b)	10-14	50/3	50/3
16.	Make bellows pockets flaps					
	a. Fold flap as indicated by marks on pattern, and stitch across flap forming a tuck (for buttonhole flap).	301	OSf-1	10-14	50/3	50/3
	b. Make two 7/8 inch cut length vertical buttonholes on each flap as indicated by marks on pattern with the inside edge of the eyelet 3/8 to 1/2 inch from the folded edge of the tuck.	Button-hole		52-56 per button-hole	30/3 or 50/3	50/3
	c. Fold flap in half with the buttonhole flap enclosed and stitch sides together.	301 or 401	SSe-2(a)	10-14	50/3 30/3	50/3 50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
16.	Make bellows pockets flaps (cont'd) d. Turn flap right side out, force out corners and edges and stitch bottom and sides 3/16 to 1/4 inch from edges. Do not catch folded edge of buttonhole flap in the stitching.	301	SSe-2(b) and OSf-1	10-14	50/3	50/3
17.	Set bellows pockets and flaps a. Turn in sides and bottom edges of the lower bellows and stitch to fronts 1/16 to 1/8 inch from edges as indicated by marks on pattern. The bellows stitching shall not be exposed beyond the finished edges of the pockets.	301	LSd-1	10-14	50/3	50/3
	b. Place two 1/2 inch vertical bartacks at tops of each pocket opening adjacent to or superimposed on the pocket setting stitching.	Bartack		28 per bartack	50/3	50/3
	c. Position flaps above pockets with top raw edge of flaps on marks as indicated on pattern.					
	d. Stitch flaps to coat 1/8 to 3/16 inch from edge. Turn flaps down and double stitch to coat, 1/4 to 5/16 inch gauge.	301	LSb1-3	10-14	50/3	50/3
18.	Make protective flap a. Fold protective flap piece in half widthwise face to face and stitch side and one end 1/4 inch from edge.	301	SSa-1	10-14	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
18.	Make protective flap (cont'd)					
	b. Turn to finished position and insert protective flap interlining.					
	c. Fold under raw edges and stitch open end 1/8 to 1/4 inch from edge.	301	SSc-1	10-14	50/3	50/3
19.	Make valve opening					
	a. Position valve opening facing face on back of left front. Stitch opening pattern in accordance with marks on pattern.	301	SSe-2(a)	10-14	50/3	50/3
	b. Cut opening close to stitching.					
	c. Turn facing to inside and edgestitch 1/16 inch from edge all sides.	301	SSe-2(b)	10-14	50/3	50/3
	d. Stitch facing to front 3/16 to 1/4 inch from raw edge.	301	LSbj-1	10-14	50/3	50/3
	e. Cut a four (4) (± 1/4) inch piece of nylon webbing. Fuse ends.					
	f. Fold under each end 1/4 inch. Position on back of front as indicated by marks on pattern and stitch 1/16 inch from folded edges.	301	LSd-1 each end	10-14	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
20.	Attach protective flap					
	a. Position protective flap on back of front as indicated by marks on pattern.					
	b. Stitch top and inner side of flap to front 1/16 inch from edge.	301	LSbj-1 each edge	10-14	50/3	50/3
21.	Make mouth hose opening					
	a. Position mouth hose opening facing on back of left front in accordance with pattern marks. Stitch a 3/4 inch diameter circle centered in the facing through both plies.	301		10-14	50/3	50/3
	b. Cut circular opening close to inner side of stitching.					
	c. Turn facing to inside and edgestitch 1/16 inch gauge.	301	SSe-2(b)	10-14	50/3	50/3
	d. Stitch facing to front 3/16 to 1/4 inch from raw edge.	301	LSbj-1	10-14	50/3	50/3
	e. Cut a 2 1/2 (± 1/4) inch piece of nylon webbing. Fuse ends.					
	f. Fold under each end 1/4 inch. Position on back of front as indicated by marks on pattern and stitch to form tunnel 1/16 inch from folded edges.	301	LSd-1 each edge	10-14	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
22.	Attach label Position label approximately 2 inches below top edge of center back and stitch four sides approximately 1/16 inch from the edges.	301		10-14	50/3	50/3
23.	Make collar a. Stitch topcollar to undercollar face to face 1/4 inch gauge. Trim corners.	301	SSe-2(a)	10-14	50/3	50/3
	b. Turn collar and work out points. Stitch outer edge and sides 3/16 to 1/4 inch from edge.	301	SSe-2(b)	10-14	50/3	50/3
24.	Make and position pencil pocket a. Hem top of pocket folding edge under 3/8 inch and stitching 1/4 inch from fold.	301	EFb-1	10-14	50/3	50/3
	b. Fold pocket as indicated by marks on patterns, turning under right side and bottom raw edges. Position on left topsleeve as indicated by marks on pattern.					
25.	Make and attach compass pockets a. Mark and stitch two (2) 3/16 inch (inside diameter) eyelets on each compass pocket piece as indicated by marks on pattern. Purling shall be on the outside.	Eyelet		22 per eyelet	50/3	50/3
	b. Hem top of pocket folding edge under 1/4 inch and stitching 1/8 inch from fold.	301	EFb-1	10-14	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
25.	Make and attach compass pockets (cont'd)					
	c. Center a 1 inch length of pile tape 1/4 inch from the top of the hem and stitch 1/16 inch from edge all sides.	301	LSbj-1 all sides	10-14	50/3	50/3
	d. Fold pocket face to face and stitch bottom and sides 1/4 (± 1/16) inch from edge.	301	SSa-1	10-14	50/3	50/3
	e. Turn pocket to finished position. Fold sides and bottom as indicated on pattern and stitch 1/16 inch from edge.	301	OSf-1	10-14	50/3	50/3
	f. Fold flap face to face as indicated by marks on pattern, turning raw edge under 1/4 inch and sandwiching top corners of pocket in between stitch sides 1/4 inch from edge.	301	SSe-2(a)	10-14	50/3	50/3
	g. Turn flap to finished position and stitch 1/8 inch from edge all sides of flap.	301	SSe-2(b) and OSf-1 and LSd-1	10-14	50/3	50/3
	h. Center a 1 inch length of hook tape on the flap facing 1/4 inch from the edge. Stitch a 1/2 by 7/8 inch rectangle through the tape and the flap with free end of the tape towards the outer portion of the flap.	301	LSbj-1	10-14	50/3	50/3
	i. Position compass pocket on left top sleeve over right side of pencil pocket and as indicated by marks on pattern.					

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
25.	Make and attach compass pockets (cont'd)					
	j. Stitch pockets to sleeve on all sides 1/16 inch from edge and across top of compass pocket through flap with flap open.	301	LSbj-1 each side	10-14	50/3	50/3
	k. Position compass pocket only on right top sleeve as indicated by marks on pattern with flap open.					
	l. Stitch pocket to sleeve, stitching all sides, 1/16 inch from edge, 3 sides and across top of pocket through flap.	301	LSbj-1 each side	10-14	50/3	50/3
26.	Make left front facing					
	a. Fold left front as indicated by marks on pattern to form facing and stitch 1/4 inch from fold. Fold under inner edge and stitch 1/16 inch from fold.	301	EFt-2	10-14	50/3	50/3
	b. Make five 7/8 inch cut length horizontal buttonholes on the under ply of the left front facing as indicated by marks on pattern with the inside edge of the eyelet 5/8 to 3/4 inch from the folded edge. Purling shall finish on the inside.	Buttonhole		52-56 per buttonhole	30/3 or 50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
27.	Join inner and outer shells					
	a. Join inner and outer shells at neckline with collar sandwiched in between commencing from left side at right front fold outershell, face to face, to form facing with collar sandwiched in between simultaneously folding inner edge of facing to finished position; fold inner shell, face to face, as indicated by marks on patterns. Stitch 1/4 inch from edges through all plies.	301	SSe-2(a)	10-14	50/3	50/3
	b. Turn collar and shells to finished position. Force out points.					
	c. Topstitch collar and tops of facings 1/16 inch from edges.	301	Similar to SSax-2(b) and SSe-2(b)	10-14	50/3	50/3
	d. Turn under bottom of left and right top fronts 3/8 to 1/2 inch. Stitch to fronts 1/16 inch from fold.	301	LSd-1	10-14	50/3	50/3
	e. Stitch along left front 2 1/4 (± 1/8) inches from edge catching all plies.	301	SSv-1	10-14	50/3	50/3
	f. Place four 1/2 inch vertical bartacks along outer row of stitching on left fronts through all plies centrally located between the buttonholes.	Bartack		28 per bartack	50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
27.	Join inner and outer shells(cont'd)					
	g. Stitch along right front facing 1/4 inch from outer edge and 1/16 inch from inner fold.	301	EFt-2 inverted	10-14	50/3	50/3
	h. Mark and stitch eleven (11) 3/16 inch (inside diameter) eyelets on the lower edge of inner back. Purling shall finish on inside of the coat.	Eyelet		22 per eyelet	50/3	50/3
	i. Join the backs with the raw edges even at the bottom and stitch with a single row of stitching 1/8 inch from edges.	301	SSa-1	10-14	50/3	50/3
	j. Join top front pieces with raw edges even and stitch with a single row of stitching 1/8 inch from edges.	301	SSa-1	10-14	50/3	50/3
	k. Center a grommet on left and right fronts through both plies, to finish in the center of the outerside of hem, approximately 3/4 inch from front edge.					
28.	Stitch sleeve and side seams					
	a. Join back arm seam of sleeves with a double stitched seam with top sleeve lapped on undersleeve.	301 or 401	LSc-2	10-14	50/3 or 30/3	50/3
	b. Stitch sleeves to coat with a double-lapped, double-stitched seam. Body of the coat shall overlap the sleeves.	301 or 401	LSc-2	10-14	50/3 or 30/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
28.	Stitch sleeve and side seams (cont'd) c. Stitch side and sleeve underarm seams in one continuous operation with a double-lapped, double-stitched seam. The fronts shall overlap the backs.	301 or 401	LSc-2	10-14	50/3 30/3	50/3 50/3
29.	Make hem a. Cut a length of elastic cord for the hem adjustment tie to finish the length of the hem between grommets plus approximately 6 inches (3 inches each side). Dip ends to prevent raveling as specified in 3.3.1.16.					
	b. Form hem turning under raw edge. Thread cord through grommets and knot.					
	c. Close hem 1/16 inch from edge enclosing the cord in the finished hem.	301	EFb-1	10-14	50/3	50/3
30.	Make and attach cuffs a. Fold cuff and stitch tab end, trim if necessary, turn, force out points and edges and edge stitch around pointed tab and along folded edge, leaving end of cuff free of topstitching to allow for insertion of sleeve into cuff.	301	SSE-2 and OSf-1	10-14	50/3	50/3
	b. Center a 7/8 inch cut length horizontal buttonhole on cuff tab with the inside edge of eyelet 1/2 to 5/8 inch from point of tab. Purling shall finish on the outside.	Buttonhole		52-56 per buttonhole	30/3 or 50/3	50/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
30.	Make and attach cuffs (cont'd)					
	c. Hem bottom of extension with the raw edge turned in. The hem shall finish 3/8 to 1/2 inch and the bottom edge shall be even with the folded edge of cuff when cuff is attached.	301	EFb-1	10-14	50/3	50/3
	d. Join underside and straight side edge of cuff to sleeve, face to face. The tabs shall face toward back.	301	LScf-2(a)	10-14	50/3	50/3
	e. Turn cuff to finished position. Turn in raw edge at top of cuff and straight side edge of cuff and, with the sleeve extension (gusset) inserted into the cuff, stitch 1/8 inch from edge.	301	LScf-2(b)	10-14	50/3	50/3
	f. Stitch tab extension to remaining raw end of sleeve extension, 1/4 inch from fold.	301	LSbk-2	10-14	50/3	50/3
	g. Place a horizontal 1/2 inch bartack at top end of cuff tabs through cuff and sleeve.	Bartack		28 per bartack	50/3	50/3
31.	Sew on buttons					
	Sew two 30-line buttons on each bellow pocket to correspond with buttonholes in pocket flaps; five on right front to correspond with buttonholes in buttonhole flap; and two on each cuff, one 1 1/4 (± 1/8) inches from straight end of cuff and the other 1 1/2 (± 1/8) inches from the first button.	301 or 101		14-16 per button 20-24 per button	30/3 30/3	30/3

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Table I - Construction of Rifleman (Type I) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
32.	Attach front and side belt loops a. Position a belt loop on each side seam as indicated by marks on patterns. Fold ends under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from each fold.	Bartack	LSd-1 each end	28 per bartack	50/3	50/3
	b. Position lower ends of front loops on fronts. Fold end under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from the fold.	Bartack	LSd-1 each end	28 per bartack	50/3	50/3
	c. Finished loop openings shall measure 3 1/2 (± 1/8) inches.					
33.	Make front eyelet Mark and stitch one (1) 3/16 inch (inside diameter) eyelet as indicated by mark on pattern on left front. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
34.	Clean coats a. Trim all thread ends. b. Remove all loose threads. c. Remove all spots, stains, and size tickets					

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Table II - Construction of Grenadier (Type II) Coat

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
	Grenadier (Type II) Coat					
1.	Cutting a. Cut the coats in strict accordance with patterns, which show size, shape, placement of pockets and flaps, warp directional lines, and notches for the proper assembling of all parts.					
2.	Replacement of damaged parts a. Replace at time of cutting any part containing damages such as a hole, smash, multiple float or slub. b. Replace any part damaged during the manufacturing process by a needle chew, cut, tear, hole, mend, burn or exposed drill hole.					
3.	Marking Mark or ticket cut parts of the coat to insure a uniform size and proper assembly throughout the coat.					
4.	Make M-16 magazine pouches a. Overedge sound separator all sides.	502 or 503	EFd-1	10-14	50/3	50/3
	b. Mark and stitch four (4) 3/16 inch (inside diameter) eyelets on each M-16 magazine pouch piece as indicated by marks on pattern. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
	c. Cut two 12 (± 1/4) inch pieces of nylon webbing. Fuse ends.					

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Make M-16 magazine pouches (cont'd)					
	d. Hem top of pouch as indicated by marks on pattern. Fold under raw edge. Stitch 1/16 to 1/8 inch from fold, simultaneously inserting the strip of nylon webbing in the hem centered with end adjacent to fold of hem.	301	SSl-1 and EFb-1	10-14	50/3	50/3
	e. Fold pouch face to face and stitch bottom sides 1/4 (± 1/16) inch from raw edge.	301	SSa-1	10-14	50/3	50/3
	f. Turn to finished position. Fold magazine pouch piece as indicated on patterns and topstitch 1/16 inch from edge.	301	OSf-1	10-14	50/3	50/3
	g. Position a 1 inch length of pile tape, centered 3/4 inch from the top of the pouch hem and stitch 1/16 inch from edge all sides.	301	LSbj-1 all sides	10-14	50/3	50/3
	h. Center a 2 1/2 inch length of 1/2 inch wide elastic webbing on the nylon webbing 3 1/2 inches from the stitched end so that the elastic webbing faces the back of the pouch material. Stitch and backstitch 1/8 to 1/4 inch from end.	301	LSbj-1	10-14	50/3	50/3
	i. Stitch sound separator to inside of pouch 3 sides catching 1/8 to 1/4 inch of the free end of the elastic webbing centered in the bottom of the pouch.	301	SSb-1	10-14	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Make M-16 magazine pouches (cont'd)					
	j. Join face to face flap facing to pouch flap simultaneously closing sides and bottom of pouch and inserting and catching 1/4 to 3/8 inch a 2 1/2 inch length of elastic webbing in the center bottom.	301	SSa-1 and SSe-2(a)	10-14	50/3	50/3
	k. Turn flap and pouch to finished position. Top-stitch flap 1/16 inch from edge 3 sides.	301	SSe-2(b)	10-14	50/3	50/3
	l. Center the end of a 12 inch piece of nylon webbing under the remaining side of the flap facing approximately 1/2 inch. Stitch side of flap to pouch 1/16 inch from edge.	301	LSd-1 and Similar to EFf-1	10-14	50/3	50/3
	m. Stitch 1/8 to 1/4 inch from edge free end of elastic webbing to underside of nylon webbing at a distance of 3 inches from stitched end of the nylon webbing.	301	SSa-1	10-14	50/3	50/3
	n. Center a 1 inch length of hook tape on the flap facing 1/4 inch from the edge. Stitch and backstitch a 1/2 by 7/8 inch rectangle through fastener and flap with free end of the fastener towards outer portion of flap.	301	LSbj-1	10-14	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
5.	Attach webbing restraint loops a. Cut two 2 (± 1/4) inch lengths of nylon webbing. Fuse ends.					
	b. Fold under each end 1/4 inch. Position on back side of fronts as indicated by marks on patterns and stitch 1/16 inch from folded edges.	301	LSd-1 each end	10-14	50/3	50/3
6.	Prepare belt loops a. Cut stripping for belt loops of sufficient width to finish 5/8 to 3/4 inch wide.					
	b. Fold stripping in half lengthwise, turn in raw edges and stitch 1/16 to 1/8 inch from each edge.	301	EFp-2	10-14	50/3	50/3
	c. If one side of stripping is selvaged, fold stripping lengthwise, turn in raw edge and stitch 1/16 to 1/8 inch from each edge.	301	EFz-2	10-14	50/3	50/3
7.	Attach front loops Position top edge of belt loop on right and left top front as indicated by marks on patterns. Fold ends under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from fold. Opening shall finish 3 1/2 (± 1/8) inches.	Bartack		28 per bartack	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
8.	Make 40 mm grenade pouches a. Mark and stitch two (2) 3/16 inch (inside diameter) eyelets on each 40 mm grenade pouch piece as indicated by marks on pattern. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
	b. Hem top of pouch, folding under raw edge and stitching 1/16 to 1/8 inch from fold. Hem shall finish 1/2 to 5/8 inch deep.	301	EFb-1	10-14	50/3	50/3
	c. Cut a 4 1/2 (± 1/8) inch length of pile tape. Taper ends to form a trapezoid with smaller length 3 1/2 (± 1/8) inches. Center longer length 1/4 inch from top of pouch hem and stitch 1/16 inch from edge all sides.	301	LSbj-1 all sides	10-14	50/3	50/3
	d. Stitch darts in pouch as indicated by marks on pattern.	301	SSa-1	10-14	50/3	50/3
	e. Position face side of pouch flap on face of pouch. Position face of flap facing on face of flap with flap facing turned under 1/4 inch sandwiching pouch in between. Stitch 1/4 to 3/8 inch from edge all sides.	301	SSa-1 and SSe-2(a)	10-14	50/3	50/3
	f. Turn flap and pouch to finished position. Topstitch flap 1/16 inch from edge all sides.	301	SSe-2(b)	10-14	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
8.	Make 40 mm grenade pouches (cont'd)					
	g. Cut a 4 1/2 (± 1/8) inch length of hook tape. Taper ends to form a trapezoid with smaller length 3 1/2 (± 1/8) inches. Center edge of shorter length 1 3/4 (± 1/8) inches from point of pouch flap facing. Stitch along 4 1/2 inch edge halfway up both sides, 1/16 inch gauge, and along center, forming a trapezoid with free end of fastener towards outer portion of flap.	301	LSbj-1	10-14	50/3	50/3
9.	Attach 40 mm grenade pouches					
	a. Prestitch bottom of pouches 1/16 inch from fold.	301	OSf-1	10-14	50/3	50/3
	b. Place 40 mm grenade pouch reinforcement pieces on backs of top fronts and backs. Prestitch as desired. Position 40 mm grenade pouches on top fronts and back as indicated by marks on patterns with flap open as follows: 3 on back 2 on right top front 2 on left top front					
	c. Stitch pouches on all sides 1/16 inch from edge, 3 sides and across top of pouch through flap.	301	LSbj-1	10-14	50/3	50/3
	d. Place two 1/2 inch vertical bartacks at the tops of each pouch at hem, 1/16 inch from sides.	Bartack		28 per bartack	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
10.	Make left top front facing Fold left top front as indicated by marks on pattern to form front facing and stitch 1/4 inch from fold.	301	SSa-1	10-14	50/3	50/3
11. ±	Attach back M-16 magazine pouches a. Stitch M-16 back reinforcement pieces to back as indicated by marks on pattern, 1/16 inch from edge, all sides.	301	LSbj-1 each edge	10-14	50/3	50/3
	b. Position four M-16 pouches with flaps open on backs as indicated by marks on pattern.					
	c. Stitch pouches on all sides, 1/16 inch from edge 3 sides and across top of pouch through flap.	301	LSbj-1 3 sides SSv-1 top	10-14	50/3	50/3
	d. Place two 1/2 inch vertical bartacks at the tops of each pouch at hem.	Bartack		28 per bartack	50/3	50/3
12.	Complete top backs a. Stitch first aid kit reinforcement piece to back side of top back as indicated by marks on pattern, 1/16 inch from edge, all sides.	301	LSbj-1 each edge	10-14	50/3	50/3
	b. Mark and stitch two (2) 3/16 inch (inside diameter) eyelets through back and reinforcement piece as indicated by marks on pattern. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
	c. Cut an 18 (± 1/2) inch piece of cotton braid. Dip ends to prevent raveling as specified in 3.3.1.16.					

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
12.	Complete top backs (cont'd)					
	d. Thread braid through eyelets with ends on the outside of back.					
	e. Place a 1/2 inch vertical bartack centrally between the eyelets, securing the braid.	Bartack		28 per bartack	50/3	50/3
	f. Fold first aid tab holder in half face to face lengthwise. Stitch sides 1/4 inch from edges.	301	SSa-1	10-14	50/3	50/3
	g. Turn face sides out, trim points.					
	h. Fold under raw edges and stitch 1/16 inch from edge.	301	SSc-1	10-14	50/3	50/3
	i. Mark and stitch two (2) 3/16 inch (inside diameter) eyelets as indicated by marks on pattern. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
	j. Fold lengthwise with eyelets on inside and stitch ends 1/8 inch from edges.	301	SSa-1	10-14	50/3	50/3
	k. With eyelets on the lower ply, place tab centrally 1/2 inch below top edge of back and stitch 1/16 inch from edge, continuing halfway down both sides.	301	SSa-1	10-14	50/3	50/3
	l. Mark and stitch four (4) 3/16 inch (inside diameter) eyelets as indicated by marks on pattern on lower portion of top back.	Eyelet		22 per eyelet	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
12.	Complete top backs (cont'd) m. Fold under bottom edge of top back 3/8 inch. Position an 8 1/2 (± 1/4) inch strip of pile fastener tape on the turn up as indicated by marks on pattern and stitch 1/16 inch from edge all sides.	301	SSs-1 and SSa-1	10-14	50/3	50/3
13.	Complete backs and attach belt loops a. Position an 8 1/2 (± 1/4) inch strip of hook fastener tape on the face side of the back as indicated by marks on pattern and stitch 1/16 inch all sides.	301	LSbj-1 all sides	10-14	50/3	50/3
	b. Position top back on back with hook and pile fastened. Stitch top back to back along bottom edge 1/8 inch from fold of top back, stopping at the fastener tape.	301	LSd-1	10-14	50/3	50/3
	c. Place a 1/2 inch vertical bartack at inner end of fastener tape through all thicknesses.	Bartack		28 per bartack	50/3	50/3
	d. Position three belt loops on backs as indicated by marks on patterns. Fold ends under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from each fold. e. Finished loop openings shall measure 3 1/2 (± 1/8) inches.	Bartack		28 per bartack	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
14.	Join shoulders Join, face to face, top fronts and fronts to backs at shoulders with a double-lapped, double stitched seam 1/4 to 5/16 inch gauge. Finished seams shall face back for outer shell and face front for inner shell.	301 or 401	LSc-2	10-14	50/3 50/3	50/3 70/2
15.	Make bellows pockets a. Mark and stitch two (2) 3/16 inch (inside diameter) eyelets on the inside bellows lower pocket, as indicated by marks on pattern. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
	b. Fold pocket in half lengthwise face to face, and stitch from top raw edge down to 5 (± 1/8) inches.	301	SSa-1	10-14	50/3	50/3
	c. Form an inverted box pleat (on outside of pocket) and stitch across bottom of pleat to hold pleat in place.	301		10-14	50/3	50/3
	d. Hem top of pocket as indicated by marks on pattern and stitch 1/16 to 1/8 inch from edge with turn under 1/4 to 3/8 inch.	301	EFb-1	10-14	50/3	50/3
	e. Stitch inside and outside bellows together face to face, 1/4 inch from the edge.	301	SSa-1	10-14	50/3	50/3
	f. Stitch pockets to bellows 1/4 inch from the edge.	301	SSe-2(a)	10-14	50/3	50/3
	g. Hem top of bellows with the raw edge turned in and the seam allowance of the pocket, turned toward bellows caught in the hem. The top edge of bellows shall be even with the top of pocket.	301	EFb-1	10-14	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
15.	Make bellows pockets (cont'd) h. Edge stitch pocket 1/16 to 1/8 inch from edge all three sides. The ends of the stitching at the top of the bellows shall be backtacked not less than 1/2 inch.	301	SSe-2(b)	10-14	50/3	50/3
16.	Make bellows pockets flaps a. Fold flap as indicated by marks on pattern, and stitch across flap forming a tuck (for buttonhole flap).	301	OSf-1	10-14	50/3	50/3
	b. Make two 7/8 inch cut length vertical buttonholes on each flap as indicated by marks on pattern with the inside edge of the eyelet 3/8 to 1/2 inch from the folded edge of the tuck.	Button-hole		52-56 per button-hole	30/3 or 50/3	50/3
	c. Fold flap in half with the buttonhole flap enclosed and stitch sides together.	301 or 401	SSe-2(a)	10-14	50/3 30/3	50/3 50/3
	d. Turn flap right side out, force out corners and edges and stitch bottom and sides 3/16 to 1/4 inch from edges. Do not catch folded edge of buttonhole flap in the stitching.	301	SSe-2(b) and OSf-1	10-14	50/3	50/3
17.	Set bellows pockets and flaps a. Turn in sides and bottom edges of the lower bellows and stitch to fronts 1/16 to 1/8 inch from edges as indicated by marks on pattern. The bellows stitching shall not be exposed beyond the finished edges of the pockets.	301	LSd-1	10-14	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
17.	Set bellows pockets and flaps (cont'd)					
	b. Place two 1/2 inch vertical bartacks at tops of each pocket opening adjacent to or superimposed on the pocket setting stitching.	Bartack		28 per bartack	50/3	50/3
	c. Position flaps above pockets with top raw edge of flaps on marks as indicated on pattern.					
	d. Stitch flaps to coat 1/8 to 3/16 inch from edge. Turn flaps down and double stitch to coat, 1/4 to 5/16 inch gauge.	301	LSb1-3	10-14	50/3	50/3
18.	Make protective flap					
	a. Fold protective flap piece in half widthwise face to face and stitch side and one end 1/4 inch from edge	301	SSa-1	10-14	50/3	50/3
	b. Turn to finished position and insert protective flap interlining.					
	c. Fold under raw edges and stitch open end 1/8 to 1/4 inch from edge.	301	SSc-1	10-14	50/3	50/3
19.	Make valve opening					
	a. Position valve opening facing face on back of left front. Stitch opening pattern in accordance with marks on pattern.	301		10-14	50/3	50/3
	b. Cut opening close to stitching.					
	c. Turn facing to inside and edgestitch 1/16 inch from edge all sides.	301	SSe-2(b)	10-14	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
19.	Make valve opening(cont'd)					
	d. Stitch facing to front 3/16 to 1/4 inch from raw edge.	301	LSbj-1	10-14	50/3	50/3
	e. Cut a four (4) ± 1/4 inch piece of nylon webbing. Fuse ends.					
	f. Fold under each end 1/4 inch. Position on back of front as indicated by marks on pattern and stitch 1/16 inch from folded edges.	301	LSd-1 each end	10-14	50/3	50/3
20.	Attach protective flap					
	a. Position protective flap on back of front as indicated by marks on pattern.					
	b. Stitch top and inner side of flap to front 1/16 inch from edge.	301	LSbj-1 each edge	10-14	50/3	50/3
21.	Make mouth hose opening					
	a. Position mouth hose opening facing on back of left front in accordance with pattern marks. Stitch a 3/4 inch diameter circle centered in the facing through both plies.	301		10-14	50/3	50/3
	b. Cut circular opening close to inner side of stitching.					
	c. Turn facing to inside and edgestitch 1/16 inch gauge.	301	SSe-2(b)	10-14	50/3	50/3
	d. Stitch facing to front 3/16 to 1/4 inch from raw edge.	301	LSbj-1	10-14	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
21.	Make mouth hose opening (cont'd) e. Cut a 2 1/2 (± 1/4) inch piece of nylon webbing. Fuse ends.					
	f. Fold under each end 1/4 inch. Position on back of front as indicated by marks on pattern and stitch to form tunnel 1/16 inch from folded edges.	301	LSd-1 each edge	10-14	50/3	50/3
22.	Attach label Position label approximately 2 inches below top edge of center back and stitch four sides approximately 1/16 inch from the edges.	301		10-14	50/3	50/3
23.	Make collar a. Stitch topcollar to undercollar face to face 1/4 inch gauge. Trim corners.	301	SSe-2(a)	10-14	50/3	50/3
	b. Turn collar and work out points. Stitch outer edge and sides 3/16 to 1/4 inch from edge.	301	SSe-2(b)	10-14	50/3	50/3
24.	Make and position pencil pocket a. Hem top of pocket folding edge under 3/8 inch and stitching 1/4 inch from fold.	301	EFb-1	10-14	50/3	50/3
	b. Fold pocket as indicated by marks on patterns, turning under right side and bottom raw edges. Position on left topsleeve as indicated by marks on pattern.					

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
25.	Make and attach compass pockets a. Mark and stitch two (2) 3/16 inch (inside diameter) eyelets on each compass pocket piece as indicated by marks on pattern. Purling shall be on the outside.	Eyelet		22 per eyelet	50/3	50/3
	b. Hem top of pocket folding edge under 1/4 inch and stitching 1/8 inch from fold.	301	EFb-1	10-14	50/3	50/3
	c. Center a 1 inch length of pile tape 1/4 inch from the top of the hem and stitch 1/16 inch from edge all sides.	301	LSbj-1 all sides	10-14	50/3	50/3
	d. Fold pocket face to face and stitch bottom and sides 1/4 (± 1/16) inch from edge.	301	SSa-1	10-14	50/3	50/3
	e. Turn pocket to finished position. Fold sides and bottom as indicated on pattern and stitch 1/16 inch from edge.	301	OSf-1	10-14	50/3	50/3
	f. Fold flap face to face as indicated by marks on pattern, turning raw edge under 1/4 inch and sandwiching top corners of pocket in between stitch sides 1/4 inch from edge.	301	SSe-2(a)	10-14	50/3	50/3
	g. Turn flap to finished position and stitch 1/8 inch from edge all sides of flap.	301	SSe-2(b) and OSF-1 and LSd-1	10-14	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
25.	Make and attach compass pockets (cont'd)					
	h. Center a 1 inch length of hook tape on the flap facing 1/4 inch from the edge. Stitch a 1/2 by 7/8 inch rectangle through the tape and the flap with free end of the tape towards the outer portion of the flap.	301	LSbj-1	10-14	50/3	50/3
	i. Position compass pocket on left topsleeve over right side of pencil pocket and as indicated by marks on pattern.					
	j. Stitch pockets to sleeve on all sides 1/16 inch from edge and across top of compass pocket through flap with flap open.	301	LSbj-1 each side	10-14	50/3	50/3
	k. Position compass pocket only on right topsleeve as indicated by marks on pattern with flap open.					
	l. Stitch pocket to sleeve, stitching all sides, 1/16 inch from edge, 3 sides and across top of pocket through flap.	301	LSbj-1 each side	10-14	50/3	50/3
26.	Make left front facing					
	a. Fold left front as indicated by marks on pattern to form facing and stitch 1/4 inch from fold. Fold under inner edge and stitch 1/16 inch from fold.	301	EFt-2	10-14	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
26.	Make left front facing (cont'd) b. Make five 7/8 inch cut length horizontal buttonholes on the under ply of the left front facing as indicated by marks on pattern with the inside edge of the eyelet 5/8 to 3/4 inch from the folded edge. Purling shall finish on the inside.	Buttonhole		52-56 per buttonhole	30/3 or 50/3	50/3
27.	Join inner and outer shells a. Join inner and outer shells at neckline with collar sandwiched in between commencing from left side, at right front fold outershell face to face to form facing with collar sandwiched in between simultaneously folding inner edge of facing to finished position; fold inner shell face to face as indicated by marks on patterns. Stitch 1/4 inch from edges through all plies.	301	SSa-1	10-14	50/3	50/3
	b. Turn collar and shells to finished position. Force out points.					
	c. Topstitch collar and tops of facings 1/16 inch from edges.	301	Similar to SSax-2(b) and SSe-2(b)	10-14	50/3	50/3
	d. Turn under bottom of left and right top fronts 3/8 to 1/2 inch. Stitch to fronts 1/16 inch from fold.	301	LSd-1	10-14	50/3	50/3
	e. Stitch along left front 2 1/4 (± 1/8) inches from edge catching all plies.	301	SSv-1	10-14	50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
27.	Join inner and outer shells (cont'd)					
	f. Place four 1/2 inch vertical bartacks along outer row of stitching on left fronts through all plies centrally located between the buttonholes.	Bartack		28 per bartack	50/3	50/3
	g. Stitch along right front facing 1/4 inch from outer edge and 1/16 inch from inner fold.	301	EFt-2 inverted	10-14	50/3	50/3
	h. Mark and stitch eleven (11) 3/16 inch (inside diameter) eyelets on the lower edge of inner back. Purling shall finish on inside of the coat.	Eyelet		22 per eyelet	50/3	50/3
	i. Prestitch backs 1/8 inch from edge on all free ends.	301	SSa-1	10-14	50/3	50/3
	j. Prestitch top fronts to fronts 1/8 inch from edges on all free ends.	301	SSa-1	10-14	50/3	50/3
	k. Center a grommet on left and right fronts through both plies, to finish in the center of the outerside of hem, approximately 3/4 inch from front edge.					
28.	Stitch sleeve and side seams					
	a. Join back arm seam of sleeves with a double stitched seam with top sleeve lapped on undersleeve.	301 or 401	LSc-2	10-14	50/3 30/3	50/3 50/3
	b. Stitch sleeves to coat with a double-lapped double-stitched seam. Body of the coat shall overlap the sleeves.	301 or 401	LSc-2	10-14	50/3 30/3	50/3 50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
28.	Stitch sleeve and side seams (cont'd) c. Stitch side and sleeve underarm seams in one continuous operation with a double-lapped, double-stitched seam. The fronts shall overlap the backs.	301 or 401	LSc-2	10-14	50/3 30/3	50/3 50/3
29.	Make hem a. Cut a length of elastic cord for the tie to finish the length of the hem between grommets plus approximately 6 inches. Dip ends to prevent raveling as specified in 3.3.1.16.					
	b. Form hem turning under raw edge. Thread cord through grommets and knot.					
	c. Close hem 1/16 inch from edge, enclosing the cord in the finished hem.	301	EFb-1	10-14	50/3	50/3
30.	Make and attach cuffs a. Fold cuff and stitch tab end, trim if necessary, turn, force out points and edges and edge stitch around pointed tab and along folded edge, leaving end of cuff free of topstitching to allow for insertion of sleeve into cuff.	301	SSe-2 and OSf-1	10-14	50/3	50/3
	b. Center a 7/8 inch cut length horizontal buttonhole on cuff tab with the inside edge of eyelet 1/2 to 5/8 inch from point of tab. Purling shall finish on the outside.	Buttonhole		52-56 per buttonhole	30/3 or 50/3	50/3

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Table II - Construction of Grenadier (Type II) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
30.	Make and attach cuffs (cont'd)					
	c. Hem bottom of extensions with the raw edge turned in. The hem shall finish 3/8 to 1/2 inch and the bottom edge shall be even with the folded edge of cuff when cuff is attached.	301	EFb-1	10-14	50/3	50/3
	d. Join underside and straight side edge of cuff to sleeve, face to face. The tabs shall face toward back.	301	LScf-2(a)	10-14	50/3	50/3
	e. Turn cuff to finished position. Turn in raw edge at top of cuff and straight side edge of cuff, and with the sleeve extension (gusset) inserted into the cuff, stitch 1/8 inch from edge.	301	LScf-2(b)	10-14	50/3	50/3
	f. Stitch tab extension to remaining raw end of sleeve extension, 1/4 inch from fold.	301	LSbk-2	10-14	50/3	50/3
	g. Place a horizontal 1/2 inch bartack at top end of cuff tabs through cuff and sleeve.	Bartack		28 per bartack	50/3	50/3
31.	Sew on buttons Sew two 30-line buttons on each bellows pocket to correspond with buttonholes in pocket flaps; five on right front to correspond with buttonholes in buttonhole flap; and two on each cuff, one 1 1/4 (± 1/8) inches from straight end of cuff and the other 1 1/2 (± 1/8) inches from the first button.	301 or 101		14-16 per button 20-24 per button	30/3	30/3

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Table II - Construction of Grenadier (Type II) Coats (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
32.	Attach front and side belt loops a. Position a belt loop on each side seam as indicated by marks on patterns. Fold ends under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from each fold.	Bartack	LSd-1 each end	28 per bartack	50/3	50/3
	b. Position lower ends of front loops on fronts. Fold end under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from the fold.	Bartack	LSd-1 each end	28 per bartack	50/3	50/3
	c. Finished loop openings shall measure 3 1/2 (± 1/8) inches.					
33.	Make front eyelet Mark and stitch one (1) 3/16 inch (inside diameter) eyelet as indicated by mark on pattern on left front. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
34.	Clean coats a. Trim all thread ends.					
	b. Remove all loose threads.					
	c. Remove all spots, stains, and size tickets.					

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Table III · Construction of Radioman (Type III) Coat

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
	Radioman (Type II) Coat					
1.	Cutting a. Cut the coats in strict accordance with patterns, which show size, shape, placement of pockets and flaps, warp directional lines, and notches for the proper assembling of all parts.					
2.	Replacement of damaged parts a. Replace at time of cutting any part containing damages such as a hole, smash, multiple float, or slub. b. Replace any part damaged during the manufacturing process by a needle chev, cut, tear, hole, mend, burn or exposed drill hole.					
3.	Marking Mark or ticket cut parts of the coat to insure a uniform size and proper assembly throughout the coat					
4.	Make M-16 magazine pouches a. Overedge sound separator all sides.	502 or 503	EFd-1	10-14	50/3	50/3
	b. Mark and stitch four (4) 1/16 inch (inside diameter) eyelets on each M-16 magazine pouch piece as indicated by marks on pattern. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
	c. Cut two 12 (± 1/4) inch pieces of nylon webbing. Fuse ends.					

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Make M-16 magazine pouches (cont'd)					
	d. Hem top of pouch as indicated by marks on pattern. Fold under raw edge. Stitch 1/16 to 1/8 inch from fold, simultaneously inserting the strip of nylon webbing in the hem centered with end adjacent to fold of hem.	301	SS1-1 and EFb-1	10-14	50/3	50/3
	e. Fold pouch face to face and stitch bottom sides 1/4 (± 1/16) inch from raw edge.	301	SSa-1	10-14	50/3	50/3
	f. Turn to finished position. Fold magazine pouch piece as indicated on patterns and topstitch 1/16 inch from edge.	301	OSf-1	10-14	50/3	50/3
	g. Position a 1 inch length of pile tape, centered 3/4 inch from the top of the pouch hem and stitch 1/16 inch from edge all sides.	301	LSbj-1 all sides	10-14	50/3	50/3
	h. Center a 2 1/2 inch length of 1/2 inch wide elastic webbing on the nylon webbing 3 1/2 inches from the stitched end so that the elastic webbing faces the back of the pouch material. Stitch and backstitch 1/8 to 1/4 inch from end.	301	LSbj-1	10-14	50/3	50/3
	i. Stitch sound separator to inside of pouch 3 sides catching 1/8 to 1/4 inch of the free end of the elastic webbing centered in the bottom of the pouch.	301	SSb-1	10-14	50/3	50/3

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Table II - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Make M-16 magazine pouches (cont'd)					
	j. Join face to face flap facing to pouch flap simultaneously closing sides and bottom of pouch and inserting and catching 1/4 to 3/8 inch a 2 1/2 inch length of elastic webbing in the center bottom.	301	SSa-1 and SSe-2(a)	10-14	50/3	50/3
	k. Turn flap and pouch to finished position. Topstitch flap 1/16 inch from edge 3 sides.	301	SSe-2(b)	10-14	50/3	50/3
	l. Center the end of a 12 inch piece of nylon webbing under the remaining side of the flap facing approximately 1/2 inch. Stitch side of flap to pouch 1/16 inch from edge.	301	LSd-1 and Similar to EFf-1	10-14	50/3	50/3
	m. Stitch 1/8 to 1/4 inch from edge free end of elastic webbing to underside of nylon webbing at a distance of 3 inches from stitched end of the nylon webbing.	301	SSa-1	10-14	50/3	50/3
	n. Center a 1 inch length of hook tape on the flap facing 1/4 inch from the edge. Stitch and backstitch a 1/2 by 7/8 inch rectangle through fastener and flap with free end of the fastener towards outer portion of flap.	301	LSbj-1	10-14	50/3	50/3
5.	Attach front M-16 magazine pouches					
	a. Stitch M-16 front pouch reinforcement pieces to top fronts as indicated by marks on patterns; 1/16 inch from edge, all sides.	301	LSbj-1 each side	10-14	50/3	50/3

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Table II - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
5.	Attach front M-16 magazine pouches (cont'd) b. Position M-16 pouches on top fronts as indicated by marks on patterns with flap open as follows: 3 on left side 1 on right side					
	c. Stitch pouches to fronts on all sides, 1/16 inch from edge, 3 sides and across top of pouch through flap.	301	LSd-1 (sides)	10-14	50/3	50/3
	d. Place two 1/2 inch vertical bartacks at the tops of each pouch at hem, 1/16 inch from sides	Bartack		28 per bartack	50/3	50/3
6.	Attach webbing restraint loops a. Cut two 2 (± 1/4) inch lengths of nylon webbing. Fuse ends.					
	b. Fold under each end 1/4 inch. Position on face side of fronts as indicated by marks on patterns and stitch 1/16 inch from folded edges.	301	LSd-1 each end	10-14	50/3	50/3
7.	Prepare belt loops a. Cut stripping for belt loops of sufficient width to finish 5/8 to 3/4 inch wide.					
	b. Fold stripping in half lengthwise, turn in raw edges and stitch 1/16 to 1/8 inch from each edge.	301	EFp-2	10-14	50/3	50/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
7.	Prepare belt loops (cont'd) c. If one side of stripping is selvaged, fold stripping lengthwise, turn in raw edge and stitch 1/16 to 1/8 inch from each edge.	301	EFz-2	10-14	50/3	50/3
8.	Attach front loops Position top edge of belt loop on right and left top front as indicated by marks on patterns. Fold ends under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from fold. Opening shall finish 3 1/2 (± 1/8) inches.	Bartack	LSd-1	28 per bartack	50/3	50/3
9.	Make left top front facing Fold left top front as indicated by marks on pattern to form front facing and stitch 1/4 inch from fold.	301	SSa-1	10-14	50/3	50/3
10.	Attach back M-16 magazine pouches a. Stitch M-16 back reinforcement pieces to back as indicated by marks on pattern, 1/16 inch from edge, all sides.	301	LSbj-1 each edge	10-14	50/3	50/3
	b. Position two M-16 pouches with flaps open on backs as indicated by marks on pattern.					
	c. Stitch pouches on all sides, 1/16 inch from edge, 3 sides and across top of pouch through flap.	301	LSbj-1 (3 sides) SSv-1 (top)	10-14	50/3	50/3
	d. Place two 1/2 inch vertical bartacks at the tops of each pouch at hem.	Bartack		28 per bartack	50/3	50/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
11.	Construct back pouch					
	a. Position face of top back on back of back. Position a strip of cotton tape along the sides and bottom. Stitch 3/8 inch from edge through all plies.	301	SSab-1	10-14	50/3	50/3
	b. Position pouch pieces face to face and stitch sides and bottom. Turn to finished position.	515 or 516 or 517 or 518	SSa-2	10-14	50/3	50/3
	c. Position top edge of pouch between top back and back. Cut two 3 1/2 inch lengths of elastic webbing. Stitch one edge of pouch to top back and other to back catching one end of elastic webbing 3/4 to 1 inch from corners. Notch corners. Turn assembly to finished position.	301	LSq-2(a) and LSak-2(a)	10-14	50/3	50/3
	d. Topstitch pouch to back and top back separately 1/16 inch from folded edge.	301	LSq-2(b) and LSak-2(b)	10-14	50/3	50/3
	e. Topstitch three remaining sides of pouch opening 1/16 inch from edge.	301	SSu shows finished seam	10-14	50/3	50/3
	f. Place a 1/2 inch diagonal bartack at upper pouch corners through all plies.	Bartack		28 per bartack	50/3	50/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
12.	Complete backs					
	a. Mark and stitch four (4) 3/16 inch (inside diameter) eyelets as indicated by marks on pattern on lower postion of top back.	Eyelet		22 per eyelet	50/3	50/3
	b. Fold under bottom edge of top back 3/8 inch. Position an 8 1/2 (± 1/4) inch strip of pile fastener tape on the turn-up as indicated by marks on pattern and stitch 1/16 inch from edge all sides.	301	SSs-1 and SSa-1	10-14	50/3	50/3
	c. Position an 8 1/2 (± 1/4) inch strip of hook fastener tape on the face side of the back as indicated by marks on pattern and stitch 1/16 inch from edge all sides.	301	LSbj-1 all sides	10-14	50/3	50/3
	d. Position top back on back with hook and pile fastened. Stitch top back to back along bottom edge 1/8 inch from fold of top back, stopping at the fastener tape.	301	LSd-1	10-14	50/3	50/3
	e. Place a 1/2 inch vertical bartack at inner end of fastener tape through all thicknesses.	Bartack		28 per bartack	50/3	50/3
13.	Position back belt loops					
	a. Position three belt loops on backs as indicated by marks on patterns. Fold ends under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from each fold.	Bartack	LSd-1 each end	28 per bartack	50/3	50/3
	b. Finished loop openings shall measure 3 1/2 (± 1/8) inches.					

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Table III- Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
14.	Join shoulders Join, face to face, top fronts and fronts to back at shoulders with a double-lapped, double-stitched seam 1/4 to 5/16 inch gauge. Finished seams shall face back for outer shell and face front for inner shell.	301 or 401	LSc-2	10-14	50/3 50/3	50/3 70/2
15.	Make bellows pockets a. Mark and stitch two (2) 3/16 inch (inside diameter) eyelets on the inside bellows lower pocket, as indicated by marks on pattern. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
	b. Fold pocket in half lengthwise face to face, and stitch from top raw edge down to 5 (± 1/8) inches.	301	SSa-1	10-14	50/3	50/3
	c. Form an inverted box pleat (on outside of pocket) and stitch across bottom of pleat to hold pleat in place.	301		10-14	50/3	50/3
	d. Hem top of pocket as indicated by marks on pattern and stitch 1/16 to 1/8 inch from edge with turn under 1/4 to 3/8 inch.	301	EFb-1	10-14	50/3	50/3
	e. Stitch inside and outside bellows together face to face, 1/4 inch from the edge.	301	SSa-1	10-141	50/3	50/3
	f. Stitch pockets to bellows 1/4 inch from the edge.	301	SSe-2(a)	10-14	50/3	50/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
15.	Make bellows pockets (cont'd)					
	g. Hem top of bellows with the raw edge turned in and the seam allowance of the pocket, turned toward bellows, caught in the hem. The top edge of bellows shall be even with the top of pocket.	301	EFb-1	10-14	50/3	50/3
	h. Edge stitch pocket 1/16 to 1/8 inch from edge all three sides. The ends of the stitching at the top of the bellows shall be backtacked not less than 1/2 inch.	301	SSe-2(b)	10-14	50/3	50/3
16.	Make bellows pockets flaps					
	a. Fold flap as indicated by marks on pattern, and stitch across flap forming a tuck (for buttonhole flap).	301	OSf-1	10-14	50/3	50/3
	b. Make two 7/8 inch cut length vertical buttonholes on each flap as indicated by marks on pattern with the inside edge of the eyelet 3/8 to 1/2 inch from the folded edge of the tuck.	Button-hole		52-56 per button-hole	30/3 or 50/3	50/3
	c. Fold flap in half with the buttonhole flap enclosed and stitch sides together.	301 or 401	SSe-2(a)	10-14	50/3 30/3	50/3 50/3
	d. Turn flap right side out, force out corners and edges and stitch bottom and sides 3/16 to 1/4 inch from edges. Do not catch folded edge of buttonhole flap in the stitching.	301	SSe-2(b) and OSf-1	10-14	50/3	50/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
17.	Set bellows pockets and flaps					
	a. Turn in sides and bottom edges of the lower bellows and stitch to fronts 1/16 to 1/8 inch from edges as indicated by marks on pattern. The bellows stitching shall not be exposed beyond the finished edges of the pockets.	301	LSd-1	10-14	50/3	50/3
	b. Place two 1/2 inch vertical bartacks at tops of each pocket opening adjacent to or superimposed on the pocket setting stitching.	Bartack		28 per bartack	50/3	50/3
	c. Position flaps above pockets with top raw edge of flaps on marks as indicated on pattern.					
	d. Stitch flaps to coat 1/8 to 3/16 inch from edge. Turn flaps down and double stitch to coat, 1/4 to 5/16 inch gauge.	301	LSb1-3	10-14	50/3	50/3
18.	Make protective flap					
	a. Fold protective flap piece in half widthwise face to face and stitch side and one end 1/4 inch from edge.	301	SSa-1	10-14	50/3	50/3
	b. Turn to finished position and insert protective flap interlining.					
	c. Fold under raw edges and stitch open end 1/8 to 1/4 inch from edge.	301	SSc-1	10-14	50/3	50/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
19.	Make valve opening					
	a. Position valve opening facing face on back of left front. Stitch opening pattern in accordance with marks on pattern.	301		10-14	50/3	50/3
	b. Cut opening close to stitching.					
	c. Turn facing to inside and edgestitch 1/16 inch from edge all sides.	301	SSe-2(b)	10-14	50/3	50/3
	d. Stitch facing to front 3/16 to 1/4 inch from raw edge.	301	LSbj-1	10-14	50/3	50/3
	e. Cut a four (4) (± 1/4) inch piece of nylon webbing. Fuse ends.					
	f. Fold under each end 1/4 inch. Position on back of front as indicated by marks on pattern and stitch 1/16 inch from folded edges.	301	LSd-1 each end	10-14	50/3	50/3
20.	Attach protective flap					
	a. Position protective flap on back of front as indicated by marks on pattern.					
	b. Stitch top and inner side of flap to front 1/16 inch from edge.	301	LSbj-1 each edge	10-14	50/3	50/3
21.	Make mouth hose opening					
	a. Position mouth hose opening facing on back of left front in accordance with marks on pattern. Stitch a 3/4 inch diameter circle centered in the facing through both plies.	301		10-14	50/3	50/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
21.	Make mouth hose opening (cont'd)					
	b. Cut circular opening close to inner side of stitching.					
	c. Turn facing to inside and edgestitch 1/16 inch gauge.	301	SSe-2(b)	10-14	50/3	50/3
	d. Stitch facing to front 3/16 to 1/4 inch from raw edge.	301	LSbj-1	10-14	50/3	50/3
	e. Cut a 2 1/2 (± 1/4) inch piece of nylon webbing. Fuse ends.					
	f. Fold under each end 1/4 inch. Position on back of front as indicated by marks on pattern and stitch to form tunnel 1/16 inch from folded edges.	301	LSd-1 each edge	10-14	50/3	50/3
22.	Attach label Position label approximately 2 inches below top edge of center back and stitch four sides approximately 1/16 inch from the edges.	301		10-14	50/3	50/3
23.	Make collar					
	a. Stitch topcollar to undercollar face to face 1/4 inch gauge. Trim corners.	301	SSe-2(a)	10-14	50/3	50/3
	b. Turn collar and work out points. Stitch outer edge and sides 3/16 to 1/4 inch from edge.	301	SSe-2(b)	10-14	50/3	50/3

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Table III - Constructionof Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
24.	Make and position pencil pocket					
	a. Hem top of pocket folding edge under 3/8 inch and stitching 1/4 inch from fold.	301	EFb-1	10-14	50/3	50/3
	b. Fold pocket as indicated by marks on patterns, turning under right side and bottom raw edges. Position on left topsleeve as indicated by marks on pattern.					
25.	Make and attach compass pockets					
	a. Mark and stitch two (2) 3/16 inch (inside diameter) eyelets on each compass pocket piece as indicated by marks on pattern. Purling shall be on the outside.	Eyelet		22 per eyelet	50/3	50/3
	b. Hem top of pocket folding edge under 1/4 inch and stitching 1/8 inch from fold.	301	EFb-1	10-14	50/3	50/3
	c. Center a 1 inch length of pile tape 1/4 inch from the top of the hem and stitch 1/16 inch from edge all sides.	301	LSbj-1 all sides	10-14	50/3	50/3
	d. Fold pocket face to face and stitch bottom and sides 1/4 (± 1/16) inch from edge.	301	SSa-1	10-14	50/3	50/3
	e. Turn pocket to finished position. Fold sides and bottom as indicated on pattern and stitch 1/16 inch from edge.	301	OSf-1	10-14	50/3	50/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
25.	Make and attach compass pockets (cont'd)					
	f. Fold flap face to face as indicated by marks on pattern, turning raw edge under 1/4 inch and sandwiching top corners of pocket in between stitch sides 1/4 inch from edge.	301	SSe-2(a)	10-14	50/3	50/3
	g. Turn flap to finished position and stitch 1/8 inch from edge all sides of flap.	301	SSe-2(b) and OSf-1 and LSd-1	10-14	50/3	50/3
	h. Center a 1 inch length of hook tape on the flap facing 1/4 inch from the edge. Stitch a 1/2 by 7/8 inch rectangle through the tape and the flap with free end of the tape towards the outer portion of the flap.	301	LSbj-1	10-14	50/3	50/3
	i. Position compass pocket on left topsleeve over right side of pencil pocket and as indicated by marks on pattern.					
	j. Stitch pockets to sleeve on all sides 1/16 inch from edge and across top of compass pocket through flap with flap open.	301	LSbj-1 each side	10-14	50/3	50/3
	k. Position compass pocket only on right topsleeve as indicated by marks on pattern with flap open.					
	l. Stitch pocket to sleeve, stitching all sides, 1/16 inch from edge, 3 sides and across top of pocket through flap.	301	LSbj-1 each side	10-14	50/3	50/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
26.	Make left front facing					
	a. Fold left front as indicated by marks on pattern to form facing and stitch 1/4 inch from fold. Fold under inner edge and stitch 1/16 inch from fold.	301	EFt-2	10-14	50/3	50/3
	b. Make five 7/8 inch cut length horizontal buttonholes on the under ply of the left front facing as indicated by marks on pattern with the inside edge of the eyelet 5/8 to 3/4 inch from the folded edge. Purling shall finish on the inside.	Buttonhole		52-56 per buttonhole	30/3 or 50/3	50/3
27.	Join inner and outer shells					
	a. Join inner and outer shells at neckline with collar sandwiched in between commencing from left side at right front fold outershell face to face to form facing with collar sandwiched in between simultaneously folding inner edge of facing to finished position; fold inner shell face to face as indicated by marks on patterns. Stitch 1/4 inch from edge through all plies.	301		10-14	50/3	50/3
	b. Turn collar and shells to finished position. Press out points.					
	c. Topstitch collar and tops of facings 1/16 inch.	301	Similar to SSax-2(b) and SSe-2(b)	10-14	50/3	50/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
27.	Join inner and outer shells (cont'd)					
	d. Turn under bottom of left and right top fronts 3/8 to 1/2 inch. Stitch to fronts 1/16 inch from fold.	301	LSd-1	10-14	50/3	50/3
	e. Stitch along left front 2 1/4 (± 1/8) inches from edge catching all plies.	301	SSv-1	10-14	50/3	50/3
	f. Place four 1/2 inch vertical bartacks along outer row of stitching on left fronts through all plies centrally located between the buttonholes.	Bartack		28 per bartack	50/3	50/3
	g. Stitch along right front facing 1/4 inch from outer edge and 1/16 inch from inner fold.	301	EFt-2 inverted	10-14	50/3	50/3
	h. Mark and stitch eleven (11) 3/16 inch (inside diameter) eyelets on the lower edge of inner back. Purling shall finish on inside of the coat.	Eyelet		22 per eyelet	50/3	50/3
	i. Prestitch backs 1/16 inch from edge all free ends.	301	SSa-1	10-14	50/3	50/3
	j. Prestitch top fronts to fronts 1/16 inch from edges all free ends.	301	SSa-1	10-14	50/3	50/3
	k. Center a grommet on left and right fronts through both plies, to finish in the center of the outerside of hem, approximately 3/4 inch from front edge.					

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
28.	Stitch sleeve and side seams					
	a. Join back arm seam of sleeves with a double stitched seam with top sleeve lapped on undersleeve	301 or 401	LSc-2	10-14	50/3 30/3	50/3 50/3
	b. Stitch sleeves to coat with a double-lapped, double-stitched seam. Body of the coat shall overlap the sleeves.	301 or 401	LSc-2	10-14	50/3 30/3	50/3 50/3
	c. Stitch side and sleeve underarm seams in one continuous operation with a double-lapped, double-stitched seam. The fronts shall overlap the backs.	301 or 401	LSc-2	10-14	50/3 30/3	50/3 50/3
29.	Make hem					
	a. Cut a length of elastic cord for the tie to finish the length of the hem between grommets plus approximately 6 inches. Dip ends to prevent raveling as specified in 3.3.1.16.					
	b. Form hem turning under raw edge. Thread cord through grommets and knot.					
	c. Close hem 1/16 inch from edge, enclosing the cord in the finished hem.	301	EFb-1	10-14	50/3	50/3
30.	Make and attach cuffs					
	a. Fold cuff and stitch tab end, trim if necessary, turn, force out points and edges and edge stitch around pointed tab and along folded edge, leaving end of cuff free of topstitching to allow for insertion of sleeve into cuff.	301	SSe-2 and OSf-1	10-14	50/3	50/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
30.	Make and attach cuffs (cont'd)					
	b. Center a 7/8 inch cut length horizontal buttonhole on cuff tab with the inside edge of eyelet 1/2 to 5/8 inch from point of tab. Purling shall finish on the outside.	Buttonhole		52-56 per buttonhole	30/3 or 50/3	50/3
	c. Hem bottom of extensions with the raw edge turned in. The hem shall finish 3/8 to 1/2 inch and the bottom edge shall be even with the folded edge of cuff when cuff is attached.	301	EFb-1	10-14	50/3	50/3
	d. Join underside and straight side edge of cuff to sleeve. The tabs shall face toward back.	301	LScf-2(a)	10-14	50/3	50/3
	e. Turn cuff to finished position. Turn in raw edge at top of cuff and straight side edge of cuff, and with the sleeve extension (gusset) inserted into the cuff, stitch 1/8 inch from edge.	301	LScf-2(b)	10-14	50/3	50/3
	f. Stitch tab extension to remaining raw end of sleeve extension, 1/4 inch from fold.					
	g. Place a horizontal 1/2 inch bartack at top end of cuff tabs through cuff and sleeve.	Bartack		28 per bartack	50/3	50/3
31.	Sew on buttons Sew two 30-line buttons on each bellows pocket to correspond with buttonholes in pocket flaps; five on right front to correspond with buttonholes in buttonhole flap; and two on each cuff, one 1 1/2 (± 1/8) inches from straight end of cuff and the other 1 1/2 (± 1/8) inches from the first button.	301 or 101		14-16 per button 20-24 per button	30/3 30/3	30/3

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Table III - Construction of Radioman (Type III) Coat (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
32.	Attach front and side belt loops a. Position a belt loop on each side seam as indicated by marks on patterns. Fold ends under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from each fold.	Bartack	LSd-1 each end	28 per bartack	50/3	50/3
	b. Position lower ends of front loops on fronts. Fold end under 1/4 inch and place a 1/2 inch horizontal bartack 1/16 to 1/8 inch from the fold.	Bartack	LSd-1 each end	28 per bartack	50/3	50/3
	c. Finished loop openings shall measure 3 1/2 (± 1/8) inches.					
33.	Make front eyelet Mark and stitch one (1) 3/16 inch (inside diameter) eyelet as indicated by mark on pattern on left front. Purling shall be on outside.	Eyelet		22 per eyelet	50/3	50/3
34.	Clean coats a. Trim all thread ends.					
	b. Remove all loose threads.					
	c. Remove all spots, stains, and size tickets.					

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3.4 Bladder

3.4.1 Materials.-

3.4.1.1 Coated cloth.- The cloth used in the construction of the bladder shall conform to type I of MIL-C-19002. The cloth used in the construction of the tape and reinforcement patches shall conform to type III of MIL-C-19002. The color shall be black.

3.4.1.2 Tape and reinforcement patches.- The tape and reinforcement patches shall be cut from the coated cloth. The tape V-stripping shall be cut in the bias direction and shall be 1 1/2 inches wide. The uncured tape shall be lightly dusted or protected by a suitable liner.

3.4.1.2.1 Dusting powder.- The dusting powder for application to the coating shall be whiting, talc or other finely divided mineral material which does not support mildew growth. Zinc stearate may be used in a blend with a mineral dusting powder, but when used, the mineral dust shall predominate.

3.4.1.3 Adhesive.- All seams, tapes, reinforcement patches and attachments shall be secured by an adhesive conforming to Class 1, 2 or 3 of MIL-A-5540 natural or black in color, except that qualification tests are not applicable.

3.4.1.4 Cord, nylon.- The nylon cord used for the spacer material shall be natural conforming to type III of MIL-C-5040. The nylon cord shall be thoroughly impreganted with polychloroprene cement before cementing to the bladder.

3.4.1.5 Thread, nylon.- The nylon thread used for stitching the two I-beam sections shall be natural conforming to type II, size E, class 2 of V-T-295.

3.4.1.6 Identification and instruction label.- Each bladder shall have a combination identification and instruction label on the inside right front conforming to type IV, classes 12 and 13 of DDD-L-20, except that all printing shall be in capitals 1/4 inch in height. The lettering shall be clearly legible after subjection to the ink adhesion test of 4.4. The label shall bear the following inscription:

BLADDER, FLOTATION
STYLE II - RADIOMAN (Example)
Name of Supplier:
Contract No.: DSA-000-000-000 (Example)
Stock No.: 4220-000-0000 (Example)

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INSTRUCTIONS:

The bladder is for use only with the Coat, Ammunition Carrying. Insert into coat through either back pocket opening or opening at back right waistline. Bladder can be inflated by pulling the lanyard to activate the CO_2 cylinder or orally by blowing into the tube provided. Use with CO_2 cylinder conforming to Type I of MIL-C-25369.

3.4.2 Mechanical inflation system.-

3.4.2.1 Holder assembly.- The CO_2 cylinder holder shall conform to dimensions shown on NAVSHIPS Drawing 805-1411524 and shall be located on the left front section facing the body when worn. The holder is designed for use with the CO_2 cylinder conforming to type I of MIL-C-25369. The holder shall be a high pressure aluminum die casting, and shall be anodized and treated with dichromate seal. The holder shall be fitted with an O-ring gasket as shown on S3306-1419352 and conforming to class B, No. 12 (AN 6227) of MIL-P-5516. The design shall be such that the cylinder will be punctured and the piercing pin will be retracted with a travel of the actuating lever through an arc of not over 90 degrees. No side pull shall be required to puncture the cylinder and retract the pin. The discharge mechanism shall operate on the eccentric principle which will allow the piercing point to return to its original position after operation even when the lever is held in the tripped position. When not in use, there shall be provision for holding the actuating lever against the side of the holder by a length of 0.016 inch diameter safety wire conforming to QQ-W-343. The lever shall be cadmium-plated, hardened steel, having a minimum Rockwell "45-N" hardness 40. The lever shall be held in place with a staked corrosion resisting steel pin. The coin set-screw shall be steel, cadmium-plated, and sodium-dichromate dipped. The holder threads shall be coated with a silicone grease after the holder has been assembled on the bladder. There shall be a cadmium-plated music wire or corrosion resisting steel spring to provide that the piercing pin separates from the cylinder immediately after puncturing to preclude any possibility of delay in inflation by the pin remaining in and sealing the hole. The piercing pin shall be a cadmium-plated hardened steel having a minimum Rockwell "45-N" hardness 50. The piercing point of the pin shall conform to the design shown on 805-1411524. The piercing pin shall be so sealed that there shall be no leakage of gas from the holder when the discharge lever is in any position. An O-ring may be provided on the shank of the piercing pin to accomplish this. The piercing pin in its normal position shall be at least 0.010 inch behind the cylinder seat to prevent accidental puncture. The holder shall be so designed that adequate clearance is provided between the open end of the holder and the CO_2 cylinder, even in the event that the seating surface, where the puncturable end of the cylinder seats, within the holder, becomes worn, so that the cylinder can be screwed into the holder to a greater depth and onto its seat.

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3.4.2.2 Holder lanyard.- The holder lanyard shall be braided and conform to the lanyard assembly specified in MIL-L-15581, except that the cord shall be 5 (± 1/4) inches long, and ends sealed by heat fusion.

3.4.2.3 Manifold stem assembly.- The holder shall be securely attached to the bladder by means of a manifold stem assembly with rubber sleeve check valve conforming to type II (62A120H1-2) of MIL-M-81385.

3.4.3 Oral inflation assembly.- The oral inflation tube and valve assembly shall conform to type II, class 1, style C (52B6586-1) of MIL-O-81375.

3.4.4 Pressure relief valve.- The pressure relief valve shall conform to Bureau of Ships Drawing No. 608-1895224, except that the valve shall relieve at 2.75 (± 0.25) psi and close at not less than 2.00 psi, the rate of flow of air shall be not lower than 4.2SCFM at 4.0 psi, and shall be located on the right front section facing the body when worn.

3.4.5 Design.- The bladders shall be yoke style with a single inflatable compartment capable of being inflated mechanically with the carbon dioxide system and orally through an oral inflation tube which is included on the bladder. A pressure relief valve is provided on the right front. All attachments face the body when bladder is worn. The shoulders shall contain inside spacer cords to insure proper air distribution. Style 1 shall have an I-beam separator inside the back portion to restrict ballooning. The uncoated side of the basic cloth shall be on the outside. All seams, V-strip and reinforcement patches shall be cemented inside the bladder.

3.4.5.1 Figure.- Figure 5 is furnished for information purposes only. To the extent of any inconsistencies between the written specifications and the figures, the written specification shall govern.

3.4.6 Patterns.- Standard patterns will be furnished by the Government. The standard shall not be altered in any way and are to be used for cutting the supplier's working patterns. The working patterns shall be identical to the standard patterns.

3.4.6.1 Pattern parts.- The component parts for the bladder shall be cut from the coated cloth in accordance with the following pattern parts:

Pattern parts	Cut parts
Style 1 - Rifleman; Grenadier	
Top ply	1
Bottom ply (face up)	1
I-beam (spacer)	2

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Pattern parts	Cut parts
Style 1 - Rifleman; Grenadier (cont'd)	
Reinforcement patches:	
Manifold	1
Oral tube	1
Relief valve	1

Template for location of "V" stripping, spacer cord, I-beam spacer, oral tube, manifold and relief valve.

Pattern parts	Cut parts
Style 2 - Radioman	
Top ply	1
Bottom ply (face up)	1
Reinforcement patches:	
Manifold	1
Oral tube	1
Relief valve	1

Template for location of "V" stripping, spacer cord, and tube, manifold and relief valve.

3.4.7 Construction.-

3.4.7.1 Use of adhesive.- In all the cementing operations, the surface to which the adhesive is to be applied shall be thoroughly cleaned with a suitable solvent in such a manner that the dusting materials (zinc stearate or talc) or any other surface contaminant are removed. Care should be exercised to insure that the coating and the base undercloth are not damaged and the adhesion between both is not impaired in any way by prolonged exposure to the solvent. The solvent used shall evaporate completely prior to the application of the adhesive and shall leave no residue. The adhesive shall be controlled to insure that old adhesive or adhesive which has partly or completely polymerized is not used. A fresh batch of adhesive shall be used at least every eight hours during the course of manufacture. Containers for the adhesive shall be free from congealed adhesive before being refilled.

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3.4.7.2 Cementing.- All seam tapes, patches, and attachments shall be cemented to the bladder without tension. The adhesive shall be applied evenly to each surface and joined. All entrapped air and wrinkles shall be removed by rolling and stretching. Cemented areas shall not contain channels, wrinkles or puckers. The openings in the patches used for reinforcing the oral inflation and manifold stem assemblies shall coincide with the openings on the attachments and shall extend not less than 1/8 inch or more than 3/8 inch beyond their edges. The ends of the seam tape shall overlap each other by approximately 3/4 inch. The adhesive, when dry or cured, shall present a neat and uniform appearance. The adhesive shall not be allowed to remain in clots nor permitted to extend in such a manner as to result in localized stiffness which might result in discomfort to the wearer, when the bladder is deflated or inflated. The adhesive, upon drying or curing, shall not cause the cloth to shrink or pucker at any point on the bladder. The adhesive shall not be allowed to run over the edges of the bladder to form stiff or hard needle points and all the edges around the entire periphery of the bladder shall be free from sharp points or ragged edges. The bladder shall be free from congealed masses of the adhesive and spot or stains resulting from excess adhesive.

3.4.7.3 Stitching.- Stitch types specified shall conform to FED-STD-751. Minimum and maximum number of stitches per inch shall be as specified.

3.4.8 Manufacturing operations requirements.- The bladder shall be manufactured in accordance with the operation requirements specified in Table IV. The supplier is not required to follow the exact sequence of operations listed provided that the finished bladders are identical to those produced by following the sequence of operations as listed in Table IV.

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Table IV - Construction of bladder

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	Cutting Cut the bladders in strict accordance with patterns which show shape, directional lines and location marks for proper assembly of attachments, spacer tubing, and I-beam.					
2.	Replacement of damaged parts a. Replace at time of cutting any coated fabric which may affect the performance of the end item. b. Replace any part damaged during the manufacturing process by a needle chew, cut, tear, hole or exposed drill hole.					
3.	Pre-cementing a. Cement all required seam joining areas and related components on the coated surface of top and bottom bladder plies in accordance with furnished templates and patterns. b. Style 1 only - Cement the coated surfaces of the two I-beam sections, 1 1/2 (± 1/16) inches down the center in the length direction, and join in accordance with the position marks on patterns.					

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Table IV - Construction of bladder (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Make I-beam (Style 1 only) a. Stitch the pre-cemented I-beam sections together in accordance with stitch location marks on patterns. b. Cement one end of the I-beam to the inner surface of the bottom ply. The width of the cementable area on each side at the end of the I-beam shall be 3/4 inch. The cementing of the I-beam shall be as indicated by template and marks on patterns. Total cemented area shall be 1 1/2 (± 1/8) inches.	301	SSa-1	8-10	E	E
5.	Attach V-stripping Cement 3/4 (± 1/8) inch wide V-stripping to the inner surface of the bottom plies around perimeters in one continuous strip, 3/8 to 1/2 inch from edge. The V-stripping may be slashed to within 1/16 inch of folded edge on curved sections. Corners shall be made square with corners of V-stripping folded over, forming a right angle.					
6.	Pressure relief valve a. Insert pressure relief valve through opening from inner surface of bottom ply, as indicated on patterns. b. Cement rubber flange of the pressure relief valve to the inner surface of the bottom ply. c. Cement pressure relief valve reinforcement patch to rubber flange and inner surface of the bottom ply. The reinforcement patch shall overlap onto the bottom ply 3/8 inch from flange edge.					

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Table IV - Construction of bladder (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
7.	Oral inflation valve assembly a. Insert oral inflation valve through opening from inner surface of bottom ply, as indicated on patterns. b. Cement rubber flange of the oral inflation valve to inner surface of the bottom ply. c. Cement oral inflation valve reinforcement patch to rubber flange and inner surface of bottom ply. The reinforcement patch shall overlap onto the bottom ply 3/8 inch from flange edge.					
8.	Manifold stem valve assembly a. Insert manifold stem valve assembly through opening from inner surface of bottom ply, as indicated on patterns. b. Cement rubber flange of manifold stem valve assembly to inner surface of the bottom ply. c. Cement manifold stem valve assembly patch to rubber flange and inner surface of bottom ply. The reinforcement patch shall overlap onto the bottom ply 1/4 inch from flange edge.					
9.	Nylon cord spacer Cement two strips of impregnated nylon cord, side by side, to the inner surface of the bottom ply in accordance with marks on the patterns. The ends of the cord shall be skived at an angle and shall measure 10 1/2 (± 1/8) inches after skiving.					

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Table IV - Construction of bladder (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
10.	Solution dusting Brush coat all areas of the inner surfaces of the top and bottom plies, that are not to be cemented together, with a dusting powder, in solution form, to prevent inside surfaces from adhering to each other.					
11.	Assemble bladder Join all cemented areas of the top and bottom bladder plies. On the style 1 bladder, the free end of the I-beam shall be cemented to the inner surface of the top ply.					
12.	Roll all seams a minimum of two rolls.					
13.	Trim bladder to size of patterns.					
14.	Remove excess cement and break all V-stripping loose.					
15.	Screw in manifold sleeve check valve.					
16.	Attach holder to manifold stem with rubber washers positioned on stem, one on each side of holder. Secure holder by tightening hex nut.					
17.	Secure oral inflation valve to connector with suitable clamp.					
18.	Turn oral inflation valve to closed position.					
19.	Secure actuating lever to holder with safety wire.					

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3.4.9 Performance requirements.- The bladder shall meet the following requirements when tested as specified in 4.4.

3.4.9.1 Operation (CO_2).- After release of a CO_2 cylinder, each segment of the bladder shall inflate in no more than 10 seconds.

3.4.9.2 Buoyancy.- The bladder shall not completely submerge when supporting a load whose weight in water is 29 pounds.

3.4.9.3 Pressure.- Inflated to a pressure of 5 psi, the bladder shall show no evidence of failure. All seams and sealed areas shall remain intact and show no sign of separation.

3.4.9.4 Leakage.- Inflated to a pressure of 2 psi, the bladder pressure shall not drop more than 0.5 psi in a 24 hour period.

3.4.9.5 Pressure relief valve.- The relief valve shall open at 2.75 ($\pm$ 0.25) pounds per square inch and close at not less than 2.00 pounds per square inch.

3.4.9.6 Adhesion.- The adhesion shall be not less than 7.5 pounds per inch width when joined coated side to coated side and not less than 5.0 pounds per inch width when joined coated side to cloth side.

3.5 Sizes and measurements.- Sizes and measurements of the finished coats and bladders shall be as shown in Tables V and VI (see figures). All measurements and tolerances are expressed in inches.

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Table V - Coats (all types)

	Short	Regular	Long	Tolerance
A. Chest				
Small	-	44 1/2	44 1/2	± 3/4
Medium	48 1/2	48 1/2	48 1/2	± 3/4
Large	52 1/2	52 1/2	52 1/2	± 3/4
B. Back length				
Small	-	31 3/8	33	± 3/4
Medium	30 1/2	31 7/8	33 1/2	± 3/4
Large	31	32 3/8	34	± 3/4
C. Sleeve length				
Small	-	24	25	± 3/4
Medium	23 1/2	24 1/2	25 1/2	± 3/4
Large	24	25	26	± 3/4

A. Chest shall be measured with coat buttoned, in line with pit of armhole from folded edge to folded edge and double the measurement.

B. Back length shall be measured along center of back from undercollar seam to bottom edge of coat.

C. Sleeve length shall be measured from top of sleeve along center of top sleeve to bottom of sleeve.

Table VI - Bladder

	Style I	Style II	Tolerance
A. Overall length	38	37	± 3/8
B. Center back length	16 1/2	14 7/8	± 1/4
C. Front panel width	9	9	± 1/4
D. Back panel width	18	10 3/8	± 1/4
E. Side panel width	-	4 3/8	± 1/4

NOTE: A, B, C, D, and E refer to figure 5. The bladder shall be laid flat and smooth in the indicated position.

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A. Measure from center of back panel to center of front panel.
B. Measure from center of back panel to center of neck opening.
C. Measure along edge of front panel.
D. Measure along edge of back panel.
E. Measure along edge of side panel.

3.6 Workmanship.- The finished coats and bladders shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection.- When required, the first article submitted in accordance with 3.2 shall be inspected for compliance with design, construction, workmanship, appearance, and dimensional requirements.

4.3 Inspection.- Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. A certificate of compliance shall be furnished for the sound separator requirements of 3.3.1.11, the dusting powder requirements of 3.4.1.2.1, and the adhesive requirements of 3.4.1.3.

4.3.2 Examination of the end item.- Examination of the end item shall be in accordance with 4.3.2.1 through 4.3.2.2.2. The applicable inspection levels and acceptable quality levels shall be as indicated in 4.3.2.3. The lot size shall be expressed in units of coats or bladders, as applicable. The sample unit shall be one finished coat or bladder, as applicable.

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4.3.2.1 Visual examination.- The coats and bladders shall be examined for defects in color, design, material, construction, workmanship and marking and the defects classified in accordance with 4.3.2.1.1 and 4.3.2.1.2.

4.3.2.1.1 Coats.-

Defect	Classification Major	Minor
MATERIAL DEFECTS AND WORKMANSHIP DAMAGES		
a. Any hole, needle chew, cut or tear	X	
b. Weakening defect such as slub, smash, float	X	
c. Any visible mend, burn or drill hole	X	
CLEANNESS		
a. Any spot or stain on outside		X
b. Thread ends not trimmed or loose threads not removed throughout the coat		X
c. Two or more size markings not removed		X
d. Any holding or basting threads visible on outside of finished coat, when applicable		X
COMPONENTS AND ASSEMBLY		
Any component part or operation omitted (unless otherwise classified herein)	X	
CUTTING		
Any part not cut in accordance with directional lines indicated on patterns or in accordance with specification requirements	X	
SEAMS AND STITCHING		
a. Accuracy of seaming		
1. Part of garment caught in any unrelated stitching operation	X	
2. Seam puckered, twisted, or pleated (unless otherwise classified herein)		X
3. End of all seams and stitchings when not caught in other seams or stitchings backtacked less than 1/4 inch		X
4. Ends of a continuous line of stitching overlapped less than 1/2 inch		X
5. Thread break not secured or stitching back of the break less than 1/2 inch		X

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Defect	Classification Major	Minor
SEAMS AND STITCHING (cont'd)		
b. Gauge of stitching		
1. Irregular, i.e., unevenly gauged, not uniform		X
2. Beyond range of width specified or varies more than 1/16 inch when no range is specified		X
c. Open seams or runoffs		
1. On all seams except bottom hem:		
-up to 1/4 inch inclusive		X
-more than 1/4 inch	X	
2. Bottom hem:		
-up to 1 inch inclusive		X
-more than 1 inch	X	
NOTE: One or more broken stitches or two or more continuous skipped or runoff stitches constitute an open seam.		
d. Raw edges		
1. On outside:		
-more than 1/4 inch to and including 1 inch		X
-more than 1 inch	X	
2. On inside and hem:		
-more than 1 inch		X
e. Seam and stitch types		
1. Not specified seam or stitch type		X
2. Looper thread on outside (when 401 stitch is used)		X
f. Stitch tension		
1. Loose tension resulting in loose seam	X	
2. Loose tension on edge or raised stitching recognized by loosely exposed loops of lower or top thread		X
3. Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	X	

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Defect	Classification Major	Minor
SEAMS AND STITCHING (cont'd)		
g. Stitches per inch (Score only when condition exists on major portion of a seam)		
1. One stitch less than the minimum or any number in excess of the maximum specified		X
2. Two or more stitches less than the minimum specified	X	
h. Seam direction		
Any seam not facing the direction specified when caught in subsequent stitching operations		X
i. Bartacks		
1. Missing, loose or misplaced	X	
2. Not size specified		X
3. Not across width of belt loops (when applicable)		X
BUTTONHOLES AND EYELETS		
a. One or more buttonholes omitted or added	X	
b. Stitches not securely caught in fabric, causing stitching to pull away from fabric	X	
c. Cut length of buttonhole not as specified	X	
d. Purling not on side specified		X
e. Eyelets not size specified		X
f. One or more broken stitches in two or more buttonholes or eyelets	X	
g. One or more broken stitches in one buttonhole or eyelet		X
h. Gimp omitted in two or more buttonholes		X
i. Ragged edges		X
j. Not specified type or uncut		X
k. Two or more eyelets omitted		X
l. Taper bar end on one or more buttonholes insecurely tacked		X
m. Buttonholes positioned less than 5/8 or more than 3/4 inch from folded edge of flap		X
n. Buttonhole flap short or tight causing puckering or twist on left front		X

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Defect	Classification Major	Minor
BUTTONS		
a. Other than size, type or style specified	X	
b. Two or more omitted or insecurely attached	X	
c. One omitted or insecurely attached		X
d. Button stitching not locked at end of stitching cycle (tug at loose end of thread, when accessible, to determine if it will ravel)		X
COLLAR		
a. Top collar tight, causing collar to curl		X
b. Fullness or puckers on top collar		X
c. Tight at joining to neck, causing puckers, or pleats on front or back		X
d. Corner of pointed end poorly shaped or not of uniform shape or size		X
e. Twisted		X
f. Points uneven in length: -more than 1/2 inch		X
g. Off center by 1/2 inch or more		X
BELLOWS POCKETS AND FLAPS		
a. Edges of pocket pleated or twisted in stitching		X
b. Pleat formed on outside of pocket	X	
c. Box pleat stitched closed on outside of pocket		X
d. Raw edge of pocket hem not turned in		X
e. Lining of flap tight, causing fullness, twisting or curling		X
f. Pocket or flap set on crookedly, or poorly shaped		X
g. Pockets or flaps out of alignment by 1/2 inch or more		X
h. Button and buttonholes not aligned, causing pocket to be pulled up, or flap to bulge or twist when buttoned		X
i. Flap not covering front or back edge of pocket by 3/16 inch or more		X
j. Not located as specified	X	

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Defect	Classification Major	Minor
SLEEVES AND CUFFS		
a. Back sleeve overlapped on front sleeve at backarm seam or sleeve lapped on body at sleeve setting seam		X
b. Hem of sleeve extension (gusset) less than 3/8 inch or more than 1/2 inch wide		X
c. Bottom of sleeve extension not even with bottom of cuff		X
d. Bartack at top of cuff not perpendicular to the stitching that stitches the tab through the tapered hem of sleeve extension		X
e. Pointed end of cuff facing toward front of sleeve	X	
f. Inside edge of buttonhole eyelet less than 1/2 inch or more than 5/8 inch from point of tab		X
g. Cut length of buttonhole more than 7/8 inch long		X
h. First button less than 1 1/8 or more than 1 3/8 inches from straight end of cuff or second button less than 1 3/8 or more than 1 5/8 inches from first button		X
i. Sleeve puckered or pleated at joining to armhole		X
j. Sleeve gusset constructed with hem on outside		X
k. Inside edge of eyelet of buttonhole less than 1/2 inch or more than 5/8 inch from pointed end of cuff		X
l. Buttonhole off center from point of cuff more than 1/8 inch		X
FRONT ALIGNMENT		
a. Fronts uneven at bottom more than 3/4 inch when buttoned		X
b. Button(s) not aligned with buttonhole(s) causing puckering or twist on front when buttoned		X
M-16 MAGAZINE POUCHES (as applicable)		
a. Sound separator missing	X	
b. Sound separator not overedged		X
c. Any pouch missing	X	
d. Nylon webbing missing or not attached as specified	X	
e. Nylon webbing ends not fused		X
f. Hook and pile fastener tape omitted or not located as specified	X	
g. Elastic webbing omitted		X
h. Nylon or elastic webbing lengths not as specified		X
i. Hook fastener tape not attached forming free end as specified		X
j. Reinforcement piece omitted		X

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Defect	Classification Major	Minor
40-MM GRENADE POUCHES (as applicable)		
a. Any pouch missing	X	
b. Hook and pile fastener tape omitted, not located as specified	X	
c. Hook fastener tape not attached forming free end as specified		X
d. Reinforcement piece omitted		X
COMPASS AND PENCIL POCKETS		
a. Any missing or not constructed as specified	X	
b. Not located as specified		X
c. Hook and pile fastener tape omitted, not located as specified	X	
d. Hook fastener tape not attached forming free end as specified		X
TAB HOLDER (FIRST AID POUCH)		
a. Omitted or not located as specified	X	
b. Tie cord omitted	X	
c. Holder not constructed as specified	X	
d. Tie cord ends not fused		X
e. Tie cord reinforcement omitted		X
f. Bartack not securing cord to coat	X	
BELT LOOPS		
a. Any loop missing or not located as specified	X	
b. Any loop twisted when set to coat	X	
c. Any loop not securely stitched to coat		X
d. Loop opening less than 3 3/8 inch	X	
e. Loop opening more than 3 5/8 inch		X
ORAL TUBE VALVE OPENING		
a. Webbing hose restraint missing	X	
b. Webbing ends not fused		X
c. Circular tube opening less than 3/4 inch in diameter	X	
d. Outer edges of facing not stitched to inner shell		X

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Defects	Classification Major	Minor
LABEL		
a. Missing, incorrect or illegible	X	
b. Not stitched on four sides		X
c. Not located as specified		X
d. Stitched through both outer and inner shells	X	
BOTTOM HEM		
a. Irregular in width by 1/4 inch or more		X
b. Grommets not located as specified		X
c. Tie cord omitted, or improper length or not attached as specified	X	
d. Twisted, puckered, or pleated by stitching		X
e. Cord not elastic	X	
f. Cord ends not tied or dipped		X
g. One or more grommets omitted, broken, bent, not securely clinched or clinched too tightly cutting surrounding fabric		X
BACK POUCH (RADIOMAN)		
a. Missing	X	
b. Inner and outer shells stitched along top	X	
c. Cotton reinforcement tape missing		X
d. Elastic webbings missing	X	
e. Either webbing piece not located 3/4 to 1 inch from corners	X	
f. Bartack missing or not located as specified	X	
BACK OPENING (RADIOMAN AND GRENADIER)		
a. Omitted	X	
b. Inner shell not constructed with half back		X
c. Fastener tapes omitted or not located as specified	X	
d. Opening less than 8 1/4 inches	X	
e. Opening more than 8 3/4 inches		X

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Defect	Classification Major	Minor
VALVE OPENING		
a. Nylon webbing missing	X	
b. Webbing length not as specified	X	
c. Webbing ends not fused		X
d. Valve opening omitted or not as specified	X	
e. Protective flap missing	X	
f. Interlining omitted	X	
g. Protective flap stitched through both shells	X	
h. Not stitched to inner shell at top and inner side only		X
i. Stitched along more than 2 edges	X	
FACINGS		
a. Twisted or full		X
b. Not constructed as specified	X	
c. Shells not attached as specified	X	

4.3.2.1.2 Bladders.-

Defect	Critical	Classification Major	Minor
MATERIAL DEFECTS AND WORKMANSHIP DAMAGES			
a. Material defective or not as specified; evidence of cuts, tears, fraying, patched areas	X		
b. Seams and V-stripping defective or not as specified; evidence of trapped air, channels, wrinkles, tape improperly applied, incomplete cementing	X		
c. Width of seams and V-stripping less than specified		X	
d. Evidence of excessively reworked air retaining seams and V-stripping		X	
e. Any clot or mass of adhesive			X
f. Any blister		X	
g. Color not uniform or as specified			X

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Defect	Classification Critical	Major	Minor
CLEANNESS			
a. One or more spots or stains			X
b. Imbedded matter which upon removal leaves an uncoated spot	X		
COMPONENTS AND ASSEMBLY			
a. Component damaged, missing, fits improperly faulty, or not as specified, impairing the function of the bladder as an item of flotation equipment	X		
b. Any required operation omitted (unless otherwise classified herein)	X		
CUTTING			
Any component part of basic material not in accordance with patterns or specification requirements	X		
HOLDER ASSEMBLY			
a. Defective, not securely attached, or not as specified	X		
b. Lanyard not securely attached to actuating lever	X		
c. Lever holding pin not staked in place	X		
d. Gage does not seat firmly against seating surface of holder	X		
e. Length of lanyard assembly not as specified			X
f. Gaskets on each side of holder attachment missing, defective, or not as specified	X		
g. Actuating lever not safety wired		X	
h. Set screw for securing CO_2 cylinder missing, defective or not as specified		X	
i. Holder threads not greased as specified		X	

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Defect	Classification Critical	Major	Minor
ORAL INFLATION ASSEMBLY			
a. Tube defective, not securely attached, or not as specified	X		
b. Valve defective or not secure in tube	X		
c. Tube shows evidence of cracks, bubbles, bulges, pinholes or other deleterious defects	X		
d. Valve not turned to closed position		X	
MANIFOLD STEM ASSEMBLY			
a. Tube defective, not securely attached or not as specified	X		
b. Not length required			X
I-BEAM (STYLE 1)			
a. Missing		X	
b. Not located as specified		X	
PRESSURE RELIEF VALVE			
a. Missing, defective, misplaced, not securely attached, or not as specified	X		
MARKING			
Missing, incorrect, illegible, misplaced, or not as specified		X	
REINFORCEMENT PATCHES			
a. Missing	X		
b. Overlap measurement less than specified		X	
METAL COMPONENTS			
Any metal surface rough or contains a nick, burr, sharp edge, or other flaw which will damage the bladder	X		

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4.3.2.2 Dimensional examination.-

4.3.2.2.1 Coats.- The coats shall be examined for defects in dimensions in accordance with the following:

a. Any measurement deviating from nominal dimensions and tolerances specified shall be scored as a measurement defect.

b. Sleeve lengths unequal in length by 1/2 inch or more shall be scored as a size measurement defect.

4.3.2.2.2 Bladders.- The bladders shall be examined for defects in dimensions. Any measurement deviating from nominal dimensions and tolerances specified shall be scored as a measurement defect.

4.3.2.3 Inspection levels and acceptable quality levels.- The inspection levels and acceptable quality levels (AQL's) expressed in defects per 100 units, for visual and dimensional examinations shall be as follows:

	AQL's	Acceptance Number	Inspection Level
For examination in 4.3.2.1.1			
Major	2.5	-	II
Total (Major and Minor combined)	15.0	-	II
For examination in 4.3.2.1.2			
Critical	-	0	I
Major	4.0	-	I
Total (Major and Minor combined)	10.0	-	I
For examination in 4.3.2.2.1	4.0	-	S-3
For examination in 4.3.2.2.2	4.0	-	S-3

4.4 Testing of the end item (bladder-.)

4.4.1 The methods of testing specified in FED-STD-601, and as listed in Table VII shall be followed. The physical values specified in Section 3 apply to the average of the determinations made on the sample unit for testing purposes, as specified in the applicable test method. The lot size shall be expressed in units of bladders. The inspection level shall be as shown in Table VII and the acceptance number shall be zero for all tests. Rejected bladders shall not be resubmitted.

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4.4.1.1 Bladder tests shall be performed by the contractor under the surveillance of the Government quality assurance representative on lots submitted for acceptance under the contract. Bladder tests shall be performed after the bladders have passed the visual examination.

4.4.1.2 In addition to the above, the contractor shall prepare a minimum of ten adhesive test samples from each day's production and shall subject the samples to the test specified in 4.4.4.6. The same cement and techniques used in the preparation of cemented seams in the bladder shall be used in the preparation of these samples.

4.4.1.3 Ink adhesion.- Ink adhesion shall be determined in accordance with the method specified in L-P-378. A minimum of three specimens shall be tested from each submitted lot of material. Any failure shall be cause for rejection of the lot.

Table VII - Testing of the bladder

Characteristic	Inspection Level	Requirement Paragraph	Test Method
Operation (CO_2)	S-3	3.4.9.1	4.4.4.1 1/
Buoyancy	S-3	3.4.9.2	4.4.4.5 1/
Pressure	I	3.4.9.3	4.4.4.2 1/
Leakage	I	3.4.9.4	4.4.4.3 1/
Pressure relief valve	I	3.4.9.5	4.4.4.4 1/

1/ One determination shall be performed per sample unit and the results reported as "pass" or "fail".

4.4.2 Atmospheric conditions.- Unless otherwise specified in the contract or order, all tests required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury (Hg) and a temperature of 77° (± 18°) F and a relative humidity of 80 percent or less. For any 24 hour period, atmospheric pressure shall vary no more than ± 1 inch of mercury (Hg) and temperature shall not vary more than ± 5° F. If the final values of the ambient temperature of barometric pressure at the end of the 24 hour air pressure tests are different from the initial values recorded at the start of the tests, the following corrections shall be made to the final pressure readings in psi:

(a) Temperature corrections.- For each degree Fahrenheit rise in temperature, 0.028 psi shall be subtracted from the final pressure reading. For each degree drop in temperature, 0.028 psi shall be added to the final pressure reading.

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(b) Barometric pressure correction.- For each 0.1 inch of Hg rise in barometric pressure, 0.049 psi shall be added to the final temperature corrected pressure reading. For each 0.1 inch of Hg drop in barometric pressure, 0.049 psi shall be subtracted from the final temperature corrected pressure reading.

(c) Pressure readings.- Pressure shall be measured by means of a Hg manometer and then coverted to psi by multiplying inches of Hg by 0.49.

4.4.3 CO_2 cylinders for test purposes.- CO_2 cylinders used in conducting the tests specified in 4.4.4.1 and 4.4.4.5 shall conform to type I of MIL-C-25369. Test cylinders need not be marked with an applicable contract number.

4.4.4 Test methods.-

4.4.4.1 Operation (CO_2).- The operation of the bladder shall be tested by releasing the CO_2 cylinder into the bladder in the normal manner with the bladder suspended by the shoulder straps over a bar having a maximum diameter of 4 inches. The bladder shall be evacuated of residual air before the CO_2 test. Each segment of the bladder shall be observed during inflation. Appreciable expansion of each segment of the bladder shall be noticeable in no more than 10 seconds after discharge of the CO_2 cylinder.

4.4.4.2 Pressure (5 psi).- With the relief valve suitably capped or sealed off, the bladder shall be inflated with air through the oral inflation valve to a pressure of 5 psi and shall be allowed to stand for a minimum of 5 minutes. The pressure drop shall not be greater than 0.4 psi. The bladder shall be examined while held under pressure. There shall be no evidence of failure of the material in any respect. All seams and sealed areas shall remain intact and show no signs of separation.

4.4.4.3 Leakage (2 psi).- The bladder shall be inflated with air through the oral inflation valve to a pressure of 2 psi and shall be allowed to stand for a minimum of 15 minutes. The pressure shall then be checked and readjusted by adding or bleeding air if necessary to the original pressure of 2 psi. The temperature and barometric pressure shall be recorded at this time. At the end of a minimum of 24 hours after the readjustment period the pressure shall be measured and corrected for any change in temperature or barometric pressure. The corrected pressure shall not be less than 1.5 psi. The oral inflation valve shall be examined while the bladder is being deflated to determine that the valve works without difficulty and shuts off securely.

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4.4.4.4 Operation of relief valve.- The bladder shall be inflated with air to test the operation of the relief valve. The valve shall relieve at 2.75 (± 0.25) psi as indicated by air bubbles while the valve is submerged. The valve shall close at not less than 2.00 psi as indicated by stoppage of air bubbles.

4.4.4.5 Buoyancy.- Immediately upon completion of the operation (CO_2) test, the pressure relief valve shall be suitably capped or sealed off. The bladder shall then be attached to a weight, the latter weighing 29 pounds when submerged in fresh, clean water at room temperature. The bladder shall support this weight (29 pounds) without complete submergence of the bladder while the weight is completely submerged.

4.4.4.6 Adhesion.- The test for simulated seam, tape, patch, and attachment adhesion shall be conducted in accordance with test method 8011 of FED-STD-601. The dimensions of the test specimens shall conform to the "strip method" requirements and the adhesion of the cemented samples shall be not less than 7.5 pounds per inch width when joined coated side to coated side and not less than 5.0 pounds per inch width when joined coated side to cloth side of the bladder material.

4.5 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing and marking complies with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery, except that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closures of container flaps, improper taping, loose strapping, or inadequate stapling; bulging or distortion of container.
Content	Number of coats or bladders per package and shipping container more or less than specified.

5. PREPARATION FOR DELIVERY

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5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Coat.- Each fully buttoned coat with size and identification label visible shall be neatly folded to measure approximately 23 inches in length by 14 1/2 inches in width. Each folded coat shall be individually packaged in a snug-fitting clear polyethylene film bag having a thickness of 0.002 inch (± 20 percent tolerance). The bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the seal made as close as possible to the open end. A 1/4 inch diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape.

5.1.1.2 Bladder.- Each bladder shall be neatly folded to measure approximately 10 1/2 inches in length by 6 1/4 inches in width. Each folded bladder shall be inserted in a clear polyethylene film bag having a thickness of 0.002 inch (± 20 percent tolerance). The polyethylene bag shall be formed with heat sealed seams that are straight, continuous and parallel to the formed edges of the bag. Closure of the polyethylene bag shall be accomplished by means of a mechanical tie (paper or plastic covered steel wire, aluminum band, etc.). Each bagged bladder shall be individually packaged in a fiberboard box conforming to style RSC, type CF, variety SW, class domestic, grade 125 of PPP-B-636. Boxes shall be constructed and closed in accordance with the requirements of PPP-B-636, except that metal fasteners shall not be used. Inside dimensions of each fiberboard box shall approximate 6 1/4 inches in length, 4 inches in width and 11 inches in depth.

5.1.2 Level C.- Coats and bladders shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.-

5.2.1.1 Coats.- Twelve coats of one description only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to class weather-resistant, grade V2s, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class weather-resistant, variety DW, grade V15c of MIL-B-17757. Level A packages shall be packed flat, front up, collars alternated front to rear, twelve in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes of one description only may be packed within the same shipping container.

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5.2.1.2 Bladders.- Twenty-four bladders of one description only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to class weather-resistant, grade V2s, style RSC of PPP-B-636. Level A packages shall be packed flat two in length, three in width, and four in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23 inches in length, 20 1/2 inches in width and 18 1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636.

5.2.2 Level B.-

5.2.2.1 Coats.- Twelve coats of one description only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to class domestic, grade 275, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, variety DW, grade 275 of MIL-B-17757. Level A packages shall be packed flat, front up, collars alternated front to rear, twelve in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes of one description only may be packed within the same shipping container.

5.2.2.2 Bladders.- Twenty-four bladders of one description only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A packages shall be packed flat two in length, three in width, and four in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23 inches in length, 20 1/2 inches in width and 18 1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.3 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with MIL-B-17757 (for coats) or PPP-B-636 (for bladders). Each shipping container shall be closed in accordance with the applicable box specifications. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C.- Coats and bladders, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.3 Marking.- In addition to any special marking required by the contract or order, interior fiberboard packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Labels, mixed sizes.- Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing and stenciling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly printed or stamped thereon and under these words shall be legibly printed or stamped the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use.- The coats and bladders are intended to be worn by male military personnel for special warfare purposes. Style I bladders are worn with the Type I and Type II coats. Style II bladder is worn with the Type III coat.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type and size of coat required (see 1.2.1).
c. Style of bladder required (see 1.2.2).
d. Whether first article sample is required (see 3.2).
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
f. When weather resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.3).

6.3 Samples and patterns.- For access to samples and patterns, address the procuring activity issuing the invitation for bids.

Custodian:
Navy - SA

Preparing activity:
Navy - SA

Project No. 8415-N554

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FIGURE 1 COAT, BUOYANT, AMMUNITION CARRYING; TYPE I - RIFLEMAN

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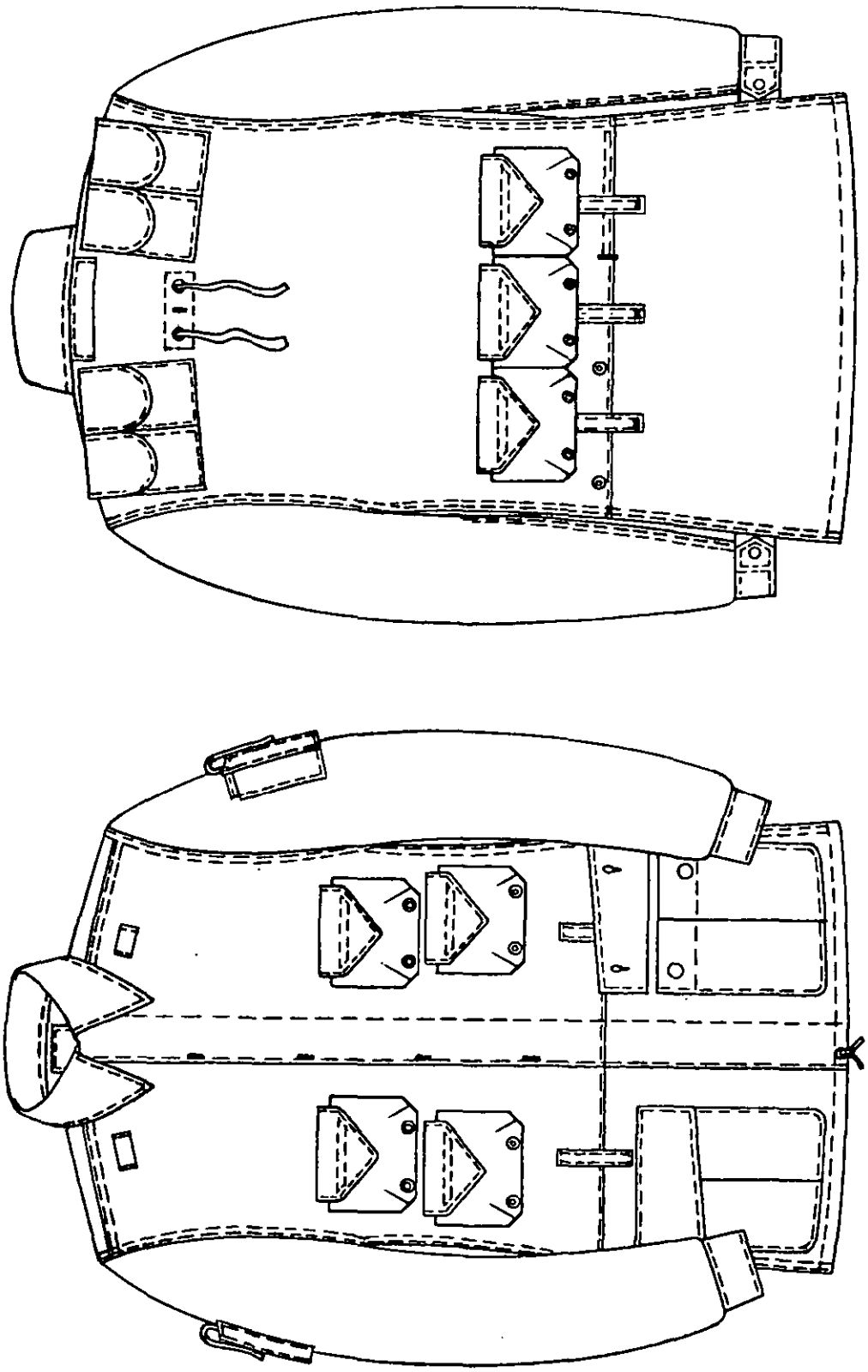
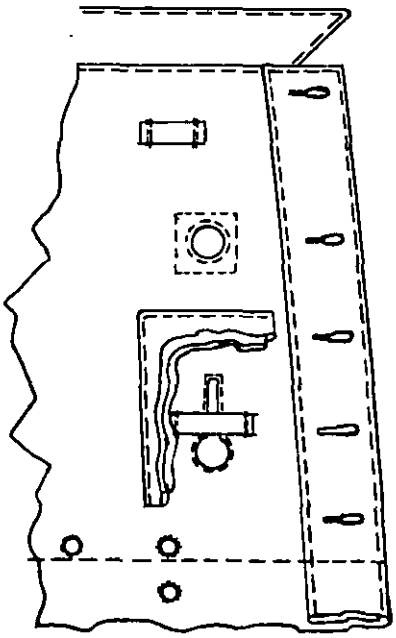


FIGURE 2 COAT, BUOYANT, AMMUNITION CARRYING; TYPE II - GRENADIER

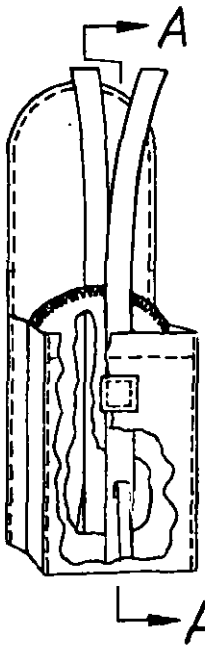
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FIGURE 3 COAT, BUOYANT, AMMUNITION CARRYING; TYPE III - RADIOMAN

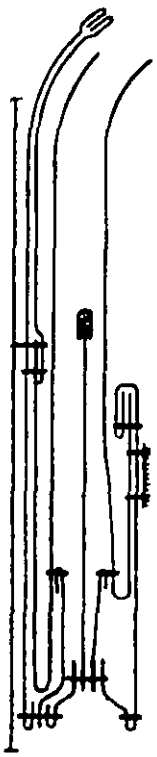
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INTERIOR DETAILS



M-16 AMMUNITION POUCH



SECTION A-A

FIGURE 4 COAT, BUOYANT, AMMUNITION CARRYING; ENLARGED VIEWS AND SECTION

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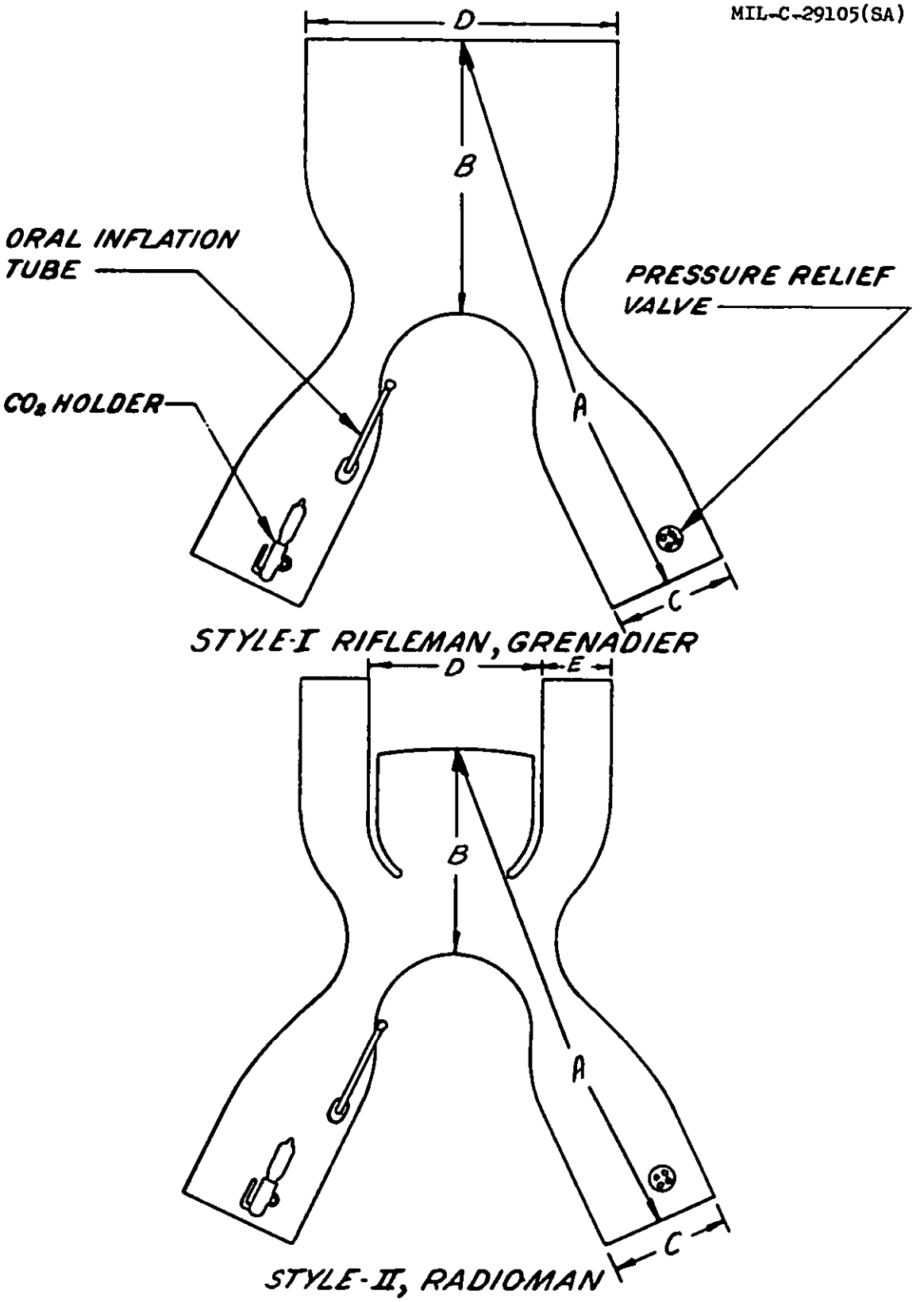


FIGURE 5 COAT, BUOYANT, AMMUNITION CARRYING; BLADDER, FLOTATION

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-29105(SA) Coats, Ammunition Carrying, Buoyant, and Bladder, Flotation

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 3ND PPSO 12502

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